



Effective Feature Selection Strategy for Supervised Classification based on an Improved Binary Aquila Optimization Algorithm

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ABSTRACT

Feature Selection (FS) is considered a crucial step in machine learning and data mining tasks, which facilitates minimizing the direct consequence of redundant and irrelevant attributes on the model's accuracy. Hence, the researchers developed various algorithms to choose the most appropriate features for improving the accuracy rate of the presented dataset. Nevertheless, these algorithms may fall into local optima problems when applied to substantial feature sizes' datasets. In this paper, for handling the FS strategy through dimensionality-lessening while improving the classification accuracy, an effective Aquila Optimization (AO) algorithm is introduced. AO has stable exploration and exploitation capabilities. It is enhanced by integrating a random position amendment approach with Local Search (LS) strategy to avoid local optima and then cloned into a binary version called Improved Binary AO (IBAO). Additionally, k-Nearest Neighbor (k-NN) and Support Vector Machine (SVM) are quality estimators. On 18 multi-scale benchmarks, the IBAO algorithm is compared with the original BAO algorithm and twelve recent algorithms, such as Binary Artificial Bee Colony (BABC), Binary Bat Algorithm (BBA), Binary Particle Swarm Optimization (BPSO), Binary Whale Optimization Algorithm (BWOA), Binary Grey Wolf Optimization (BGWO), Binary Sailfish Optimizer (BSFO), Binary Henry Gas Solubility Optimization (BHGSO), Binary Harris Hawks Optimization (BHHO). According to Wilcoxon's rank-sum test ($\alpha = 0.05$), the dominance and significant influence of IBAO are apparent on both small and large dimensional benchmarks and obtained classification accuracy up to 100% in some of these benchmarks integrated with a feature reduction size down to 92%.

1. Introduction

The fundamental goal of classification is to determine the class of each instance in a dataset based on training as the first phase and testing as the second phase. The first phase involves developing a classifier model to decide which class each instance in the training set belongs to (records with available classes). The classifier's accuracy is determined in the second phase using a set of testing examples (the classifier has not previously trained on them, but their associated classes are known to us).

Dimensionality is a crucial challenge that might degrade the classification process' performance. Several real-world applications and areas, such as medical diagnosis (Georges et al., 2020; Peng, Zhu, Wang, An, & Shen, 2019), bioinformatics (Awada, Khoshgoftaar, Dittman, Wald, & Napolitano, 2012; Urbanowicz, Olson, Schmitt, Meeker, & Moore, 2018), fault detection (Belaout, Krim, Mellit, Talbi, & Arabi, 2018), text mining (Manochandar & Punniyamoorthy, 2018), gene expression array analysis (Li & Wong, 2002), text categorization (Yang, Wu,

Deng, Zhang, & Yang, 2002), genomics (Kohane, Butte, & Kho, 2002), cancer detection (Prabukumar, Agilandeewari, & Sangaiah, 2017), image classification (Ibrahim, Abd Elaziz, Ewees, Selim, & Lu, 2018), computer vision (Arigbabu, Liao, Abdullah, & Noor, 2016), signal processing (Kung, Luo, & Mak, 2010), image retrieval (Aziz, Ewees, & Hassanien, 2018), and others, rely on high-dimensional datasets with some of features that are useless, redundant, or noisy (Yu & Liu, 2003). The presence of that features in the dataset may induce data over-fitting or cause the learning system to become confused (Liu & Motoda, 1998). Feature Selection (FS) is utilized as an essential preprocessing step to address this problem, which facilitates limiting the detrimental impact of ill-suited features on the classification model performance. As the objective of FS is to select a (close) subset of optimal features that have a significant influence on the classification model performance and accuracy and discard the others that have either no effect or negative effect on the model with a minimum cost (finding the near-optimal solution) (Faris et al., 2018). Various algorithms for determining the

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essential features have been developed in studies to extract meaningful information that enhances overall classification performance and accuracy for a given dataset. In actuality, conventional algorithms struggle to identify the significant set of features for a dataset with a large number of features.

FS algorithms could be categorized into three sets: filter (Barddal, Enembreck, Gomes, Bifet, & Pfahringer, 2019), wrapper (González, Ortega, Damas, Martín-Smith, & Gan, 2019), and embedding (Zhang, Nie, Li, & Wei, 2019). Filter algorithms deal with the FS process and classifier model as separate phases. In the first phase, The FS process depends on specific metrics to extract features that significantly impact the classification process from the dataset and disregard the others. Following that, only selected attributes are passed to the classification model to start the classification phase as a second phase. On the other hand, the wrapper algorithms dynamically adjust the selected subsets of features based on classifier accuracy. FS often employs the wrapper approach to generate several subsets of features based on specific search approaches and then estimates their relevance by running a classification algorithm. FS adjoins with a classifier for the embedded algorithms to determine which features should be retained or discarded from the dataset.

According to Zhigljavsky (2012), FS is arguably an NP-complete combinatorial optimization issue; if the number of dataset features is x , there will be 2^x various solutions, and it is computationally expensive to obtain the near-optimal feature subset from them. Furthermore, according to Amaldi and Kann (1998) and Khurma, Aljarah, and Sharieh (2021), FS can also be regarded as an NP-hard optimization issue with an exponential relationship between the computational time and the problem complexity. Consequently, meta-heuristic (MH) algorithms have garnered considerable attention (Rodrigues, Yang, Souza, & Papa, 2015). Mainly due to their superior performance in tackling various optimization scenarios. Swarm intelligence (SI) algorithms, Evolutionary Algorithms (EA), Physics-based algorithms (PhA), and Human-based algorithms are the four types of MH algorithms.

SI algorithms are a collection drawn from swarms and animal behavioral patterns (Dorigo, Maniezzo, & Coloni, 1996; Eberhart & Kennedy, 1995; Karaboga & Basturk, 2007). Particle Swarm Optimization (PSO), inspired by the natural behaviors of swarm objects and in which each object represents a possible solution, is one of the most utilized SI algorithms (Eberhart & Kennedy, 1995). The foraging activities of several ant species promoted Ant Colony Optimization (ACO) (Dorigo et al., 1996). In nature, ants release pheromones on the ground to indicate the best route for the colony members to take (Dorigo et al., 1996). It has garnered considerable attention and has been employed in various optimization applications, particularly in vehicle routing issues and FS (Abualigah & Diabat, 2021). The honey bee colony's habits prompted the Artificial Bee Colony (ABC) creation. It has three different groups of bees: employed bees seeking food resources, onlooker bees picking food sources, and scout bees searching for food sources at random (Karaboga & Basturk, 2007). Over and above, several SI algorithms have been developed in the literature. They have achieved higher performance in a wide range of optimization problems, such as Salp Swarm Algorithm (SSA) (Mirjalili et al., 2017), Cheetah Based Optimization Algorithm (Klein, Mariani, & dos Santos Coelho, 2018), Migrating Birds Optimization (Tongur, Ertunc, & Uyan, 2020), Owl Optimization Algorithm (de Vasconcelos Segundo, Mariani, & dos Santos Coelho, 2019), Modified Social-Spider Optimization Algorithm (Klein, Segundo, Mariani, & Coelho, 2015), Bacterial Foraging Optimization Algorithm (Das, Biswas, Dasgupta, & Abraham, 2009), Chaotic Coyote Algorithm (Pierezan et al., 2021), Emperor Penguin Optimizer (EPO) (Dhiman & Kumar, 2018) and Marine Predators Algorithm (MPA) (Faramarzi, Heidarinejad, Mirjalili, & Gandomi, 2020). The EA algorithms are developed by emulating biological evolutionary behaviors such as mutation, crossover, and selection. Genetic Algorithm (GA) (Holland, 1992), Differential Evolution (DE) (Storn & Price, 1997), Biogeography-Based Optimizer (BBO) (Simon, 2008) and

Invasive Tumor Growth Optimizer (ITGO) (Tang, Dong, Jiang, Li, & Huang, 2015) are notable EA-based MH algorithms that have proven their efficiency in numerous optimization applications such as face recognition, network anomaly detection, parallel machine scheduling, data mining, feature selection (Tang et al., 2015).

PhA algorithms are inspired by physical laws (Mirjalili, Mirjalili, & Hatamlou, 2016; Rashedi, Nezamabadi-Pour, & Saryazdi, 2009). The most prevalent PhA algorithms are Big Bang-Big Crunch (BBBC) (Erol & Eksin, 2006), Gravitational Search Algorithm (GSA) (Rashedi et al., 2009), Multi-verse Optimizer (MVO) (Mirjalili et al., 2016). Also, they received wide attention and have been utilized to enhance and solve various applications and problems. Examples of such issues include image segmentation, feature selection, engineering designs, global optimization problems, time series forecasting, and others. In Human-based procedures, Researchers suggested multiple MH algorithms for tackling optimization issues by modeling some human like behaviors, such as Teaching based learning algorithm (TBLA) (Rao, Savsani, & Vakharia, 2012), which was proposed by Rao and Patel (2013) to solve constrained optimization problems, Socio Evolution Learning Optimization Algorithm (SELOA) (Kumar, Kulkarni, & Satapathy, 2018), sine cosine algorithm (Mirjalili et al., 2016), and volleyball Premier League Algorithm (Moghdani & Salimifard, 2018).

All the MH algorithms mentioned earlier have common characteristics. As their optimization process consists of two phases, which are exploration and exploitation (Abualigah, Diabat, & Geem, 2020; Salcedo-Sanz, 2016). The MH algorithms create random operators to investigate different problem space areas in the exploration phase. The optimization algorithm subsequently attempts to reach the best solution from the problem space in the exploitation phase. An efficient MH optimization algorithm must balance the exploration and exploitation phase inclinations to avoid entrapment at local optima.

There have been several recent comprehensive surveys describing the issues that emerge in feature selection in many disciplines of machine learning (Chandrashekar & Sahin, 2014; Dhal & Azad, 2021; Miao & Niu, 2016). These also extensively illustrate the FS process's difficulty and how dimensionality reduction may be implemented for diverse datasets. These concepts are also briefly mentioned in the current paper's introduction section. Another research that surveys the same subject area is Xue, Zhang, Browne, and Yao (2015). This research concentrates on evolutionary computation methodologies to reach the aim, making it more relevant to the present work. Brezočnik, Fister, and Podgorelec (2018) introduces an overview of FS research focusing on methods and techniques using swarm intelligence systems.

Abd El-Mageed, Gad, Sallam, Munasinghe, and Abohany (2022) proposed a novel binary version of the adaptive wind-driven optimization algorithm for handling FS in the supervised classification tasks incorporated with the simulated annealing method and crossover strategy to improve its ability to explore for feasible regions and exploit within these regions sequentially. Satisfactory results had been achieved, revealing how the performance and capability of their approach. Gad, Sallam, Chakraborty, Ryan, and Abohany (2022) presented a novel binary version of the sparrow search algorithm to handle FS in the supervised classification tasks combined with random repositioning of roaming agents as well as the LS method, which would struggle to choose the optimum or near-optimum attributes subset from a given dataset and keeping the accuracy rate maximized. Abdelkader, Gad, Abohany, and Sorour (2022) suggested a framework for evaluating student satisfaction levels with the online learning process during the COVID-19 lockdown. FS was typically used to choose the minimum feature size. Therefore, the FS process greatly enhanced the classification accuracy of their proposed model.

Zivkovic et al. (2022) introduced a new salp swarm algorithm (SSA) algorithm that overcomes shortcomings within the original one (Mirjalili et al., 2017). A novel SSA meta-heuristic algorithm (SSARM-SCA) was constructed by merging the original SSA algorithm and sine cosine

algorithm (SCA) meta-heuristics with a guided replacement mechanism. The suggested SSARM-SCA was first tested and assessed on challenging samples of functions with 30 attributes from the CEC2013 benchmark suite. Besides that, it was implemented as a wrapper-based feature selection technique and verified against 21 well-known benchmark datasets retrieved from the University of California, Irvine (UCI) repository (Dua, Graff, et al., 2017). Wu et al. (2022) developed an efficient Teaching-Learning-Based Optimization (ETLBO) to increase classification accuracy by adjusting the SVM and Tsallis entropy parameters. The adaptive weight method and Kent chaotic map were utilized to improve the traditional TLBO's optimum ability. The presented approach attempted to overcome the original TLBO's fundamental limitations, which include stranded in local optimal and an imbalance between the search methods. Experiments based on 16 classical datasets were considered to assess the ETLBO's performance, and the outcomes were evaluated to those of other optimization techniques. The acquired results demonstrated that the proposed ETLBO outperforms different approaches in classification accuracy and optimal ability in almost all tested tasks. In Kareem, Mostafa, Hashim, and El-Bakry (2022), the GTO-BSA technique was acquired to propose a special and unique FS strategy by enhancing the efficiency of the Gorilla Troops Optimizer (GTO) based on the algorithm for bird swarms (BSA). BSA was integrated with GTO to improve the performance exploitation of the Original GTO. The proposed GTO-BSA had a higher potential to locate feasible spaces with optimal solutions. GTO-BSA's performance was evaluated compared to other competing algorithms on four IoT-IDS datasets.

An enhanced version of the Aquila algorithm was proposed in Gao, Shi, Xu, and Xu (2022) that enhanced the basic variant of the Aquila algorithm in three steps. Initially, they improved the optimization process by introducing the search control factor to enhance hunting strategies. Secondly, they added the random opposition-based learning approach to improve the exploitation capability. Lastly, the Gaussian mutation approach was utilized to enhance the exploration stage. Their proposed methodology was tested on CEC2019 test functions and 23 benchmarks to estimate the optimization ability. The empirical outcomes confirmed the majority of their proposed method than others. In Yu, Jia, Zhou, and Hussien (2022), three different strategies were added to the original version Aquila algorithm: chaotic local search, opposition-based learning, and restart strategy. Their proposed method was tested utilizing five engineering-constrained problems and 29 CEC 2017 functions. The experimental outcomes confirmed the efficiency and superiority of their method in tackling multiple optimization problems.

As previously stated, adopting meta-heuristic algorithms can provide significant advantages, and therefore the FS problem will be dealt with by recent approaches. The theorem of "No Free Lunch" (Wolpert & Macready, 1997) can be a sufficient response to this inquiry since it demonstrates that no unique method can handle all optimization problems, so it can perform efficiently on a benchmark while not working well with others. Accordingly, the choice of a reliable optimization approach for addressing the FS in supervised classification tasks remains open for researchers.

1.1. Motivations

The AO is a new meta-heuristic optimization algorithm inspired by nature, which was suggested by Abualigah et al. (2021) to handle optimization problems with continuous behavior. This paper presents an enhanced binary variant of AO, called the IBAO algorithm, to address the FS in supervised classification problems motivated by multiple improvements. Firstly, the proposed algorithm has the potential to achieve adequate search accuracy, high convergence speed, and increased stability, as well as avoid local optima, which have not been systematically implemented for the FS problem. Secondly, compared to state-of-art well-developed meta-heuristic methods and classical optimizers, the

proposed IBAO achieves better efficacy by getting optimum or near-optimum results for many of the considered problems. In addition, IBAO obtains discrete values from continuous ones based on a transfer function and an integrated random position amendment approach and LS strategy for enhancing exploration capability and improving the optimal results achieved within the solution space through iterations. To confirm the potency of this algorithm for picking the most related features, two standard machine learning classification algorithms, k -Nearest Neighbor (k -NN) and Support Vector Machine (SVM), are utilized to calculate average classification accuracy across 30 separate implementations of the proposed algorithm on 18 benchmarks with multi-scale features and instances chosen from the UCI machine learning repository (Frank, 2010).

1.2. Contributions

This paper proposed a novel algorithm for handling FS strategy using the AO algorithm; its fundamental contributions can be explained in the following:

1. AO is modified and reproduced to develop a binary version called the IBAO algorithm.
2. The IBAO algorithm incorporates a random position amendment approach and a revolutionary Local Search (LS) strategy.
3. IBAO algorithm is used for wrapper FS in supervised classification tasks.
4. The outcomes are assessed using various performance metrics involving mean fitness rate, mean accuracy rate, and the mean number of features selected.
5. The impact of the suggested IBAO algorithm using the two presented machine classification algorithms k -NN and SVM, is verified in comparison with its peers utilizing a Wilcoxon's non-parametric rank-sum test (5% significance level).

1.3. Structure

The following is the structure of this paper. Section 2 provides an explanation of the original AO and a brief discussion on the two presented classification algorithms used in this paper: k -NN, and SVM; Section 3, investigates and elaborates for the suggested IBAO algorithm to handle FS strategy; Section 4 presents the experimental outcomes and comparisons with some state-of-art methods, and the conclusion as well as points for future research in Section 5.

2. Fundamental preliminaries

2.1. Aquila Optimization (AO) algorithm

To imitate the Aquila's natural behaviors throughout the prey's hunting process, an efficient nature-inspired meta-heuristic optimization algorithm named AO algorithm (Abualigah et al., 2021) was presented.

The Accipitridae family includes the Aquila, the most famous and researched prey bird in the world because of its courage in hunting. It is a deep brown bird with golden-brown plumage on the neck's back and lives in large nests high in the mountains. Aquila can hunt diverse animals, like squirrels, rabbits, and marmots, depending on its many capabilities, the most important of which are agility, velocity, solid and enormous feet, and pointed claws (Steenhof, Kochert, & McDonald, 1997).

Aquila is also known for its capacity to transition between hunting tactics smoothly and deftly, depending on the situation. In general, Aquila depends on four tactics to capture various prey. These four tactics can be combined mathematically in two stages: exploration and exploitation. The following subsections present a brief explanation and the mathematical formula of these two stages.

2.1.1. Exploration stage

The first tactic in the exploration stage is called **flying highly with descending vertically**. The Aquila seeks appropriate prey by flying high above the land. When the best prey's region is specified, the Aquila spreads its wings and tail and descends vertically towards the prey by commencing a long, low-angled glide and increasing its speed. Mathematically, this tactic is called the **extended exploration** and can be formulated as follows:

$$X_{E-expl}^{t+1} = X_{Best}^t \times \left(1 - \frac{t}{T}\right) + (X_{Av}^t - X_{Best}^t * rand), \quad (1)$$

$$X_{Av}^t = \frac{\sum_{i=1}^N X_i^t}{N}. \quad (2)$$

where X_{E-expl}^{t+1} represents Aquila's next updated position at the next $t+1^{th}$ iteration; it is created by the extended exploration tactic (first tactic). X_{Best}^t indicates the current best position discovered while searching at t th iteration, which refers to the approximate position of the prey. t presents the current iteration, while T is the maximum number of allowed iterations. To dominate the extended exploration across the set of iterations, the expression $(1 - \frac{t}{T})$ is utilized. X_{Av}^t denotes the mean value of the present positions at the current t th iteration, which is evaluated through Eq. (2). $rand$ is evenly dispersed arbitrary number within the range $[0, 1]$. X_i^t denotes the position discovered while searching at t th iteration, and N is the overall number of candidate positions (size of population).

The second tactic in the exploration stage is defined as **flying contour with a short sliding attacking**. In this tactic, the Aquila carefully examines the selected region of the target prey, whether the prey is in a running situation or a flying situation from a high altitude. The Aquila hovers over the picked prey, and in preparation for the attack, it ascends at a low height over the ground. Mathematically, this tactic is called the **limited exploration**, and Aquila's updated position is defined as follows:

$$X_{L-expl}^{t+1} = X_{Best}^t \times Levy(D) + X_R^t + (y - z) * rand, \quad (3)$$

$$Levy(D) = s \times \frac{\mu \times \sigma}{|v|^{\frac{1}{\beta}}}, \quad \sigma = \left(\frac{\Gamma(1 + \beta) \times \sin(\frac{\pi\beta}{2})}{\Gamma(\frac{1+\beta}{2}) \times \beta \times 2^{(\frac{\beta-1}{2})}} \right), \quad (4)$$

$$y = (r_1 + U \times D_1) \times \cos(-\omega \times D_1 + (\frac{3 \times \pi}{2})), \quad (5)$$

$$z = (r_1 + U \times D_1) \times \sin(-\omega \times D_1 + (\frac{3 \times \pi}{2})). \quad (6)$$

where X_{L-expl}^{t+1} stands for Aquila's next updated position at the next $t+1^{th}$ iteration, as determined by the limited exploration tactic (second tactic). X_{Best}^t indicates the current best position discovered while searching at t th iteration, $Levy(D)$ is the levy flight distribution function, derived using Eq. (4), D is the problem's dimensional space, X_R^t is a randomly chosen Aquila's position at t th iteration. The twisting form in the search is represented by the values y and z , which are evaluated using Eqs. (5) and (6), respectively, $rand$ is an arbitrary number within the range $[0, 1]$.

s is a scale factor whose value is defined by the scales of the problem ($s = 0.01$). μ and v are arbitrary numbers distributed equally within the range $[0, 1]$, σ is calculated by Eq. (4), where Γ represents the gamma function, $\beta = 1.5$, and Pi is the pi function. r_1 indicates the number of search rotations ranges from 1 to 20 for specifying the number of search rotations, and U is a small value set to 0.00565. D_1 is an integer number from 1 to the length of the problem's dimensional space D , and ω is a small value set to 0.005.

In the exploration stage, when $t \leq (\frac{2}{3}) * T$ is true, where T and t are the maximum number of iterations and the present iteration, respectively, the exploration stage can be expressed mathematically, for making the exploration's quality more efficient, as follows:

$$X_{Expl}^{t+1} = \begin{cases} X_{E-expl}^{t+1}, & \text{if } rand < 0.5, \\ X_{L-expl}^{t+1}, & \text{if } rand \geq 0.5, \end{cases} \quad \text{if } t \leq (\frac{2}{3}) * T. \quad (7)$$

where X_{E-expl}^{t+1} represents Aquila's next updated position at the next $t+1^{th}$ iteration; it is created by the extended exploration tactic (first tactic). X_{L-expl}^{t+1} stands for Aquila's next updated position at the next $t+1^{th}$ iteration, as determined by the limited exploration tactic (second tactic). t means the present iteration, while T is the maximum number of iterations.

2.1.2. Exploitation stage

The first tactic in the exploitation stage is called **flying low with descending slowly**. The Aquila identifies and exploits the intended prey's region precisely in this tactic, then descends to the ground and gradually approaches the prey for assault. Sluggish prey or prey that does not have an escape response is the species in which this hunting tactic is used. This tactic is known as **extended exploitation** mathematically, and it may be expressed as follows:

$$X_{E-exploit}^{t+1} = (X_{Best}^t - X_{Av}^t) \times \alpha - rand + ((UB - LB) \times rand + LB) \times \delta. \quad (8)$$

where $X_{E-exploit}^{t+1}$ is Aquila's next updated position at the next $t+1^{th}$ iteration, as determined using the extended exploitation tactic (first tactic). X_{Best}^t refers to the current best position obtained during the entire search at t th iteration. X_{Av}^t is the mean value of the current positions at t th iteration and can be assessed through Eq. (2). α and δ are adjustment parameters for the exploitation stage, which are set in this paper to a small value (0.1). $rand$ is a random number within the range $[0, 1]$. UB and LB denote the upper bound and the lower bound for the space of search, respectively.

The second tactic in the exploitation stage is known as **walking on the ground and capturing prey**, where the Aquila descends to the land, approaches the prey, tracks it, and then attacks the prey over the ground. This tactic is called **limited exploitation** mathematically, and it can be described as follows:

$$X_{L-exploit}^{t+1} = QF \times X_{Best}^t - (G_1 \times X^t \times rand) - G_2 \times Levy(D) + rand \times G_1, \quad (9)$$

$$QF^t = t^{\frac{(2 \times rand - 1)}{(1-T)^2}}, \quad (10)$$

$$G_1 = 2 \times rand - 1, \quad G_2 = 2 \times \left(1 - \frac{t}{T}\right). \quad (11)$$

where $X_{L-exploit}^{t+1}$ is Aquila's next updated position at the next $t+1^{th}$ iteration, as generated by the limited exploitation tactic (second tactic). The value of quality function QF is used to balance the search strategy and is estimated using Eq. (10). X_{Best}^t means the current best position obtained during the entire search at t th iteration. G_1 reflects Aquila's random movements while tracking and hunting the prey, as created by Eq. (11). X^t is the present position at the t th iteration. $rand$ is a random value within $[0, 1]$. The flying slope of Aquila when following the prey is denoted by G_2 , which has decreasing values from 2 to 0 and is calculated using Eq. (11). $Levy(D)$ is the levy flight distribution function, derived using Eq. (4). T and t are the maximum number of iterations and the present iteration, respectively.

In the exploitation stage, when $t > (\frac{2}{3}) * T$ is true, $X_{Exploit}$ is adjusted to improve the exploitation's performance using the following formula:

$$X_{Exploit}^{t+1} = \begin{cases} X_{E-exploit}^{t+1}, & \text{if } rand < 0.5, \\ X_{L-exploit}^{t+1}, & \text{if } rand \geq 0.5, \end{cases} \quad \text{if } t > (\frac{2}{3}) * T. \quad (12)$$

where $X_{E-exploit}^{t+1}$ is Aquila's next updated position at the next $t+1^{th}$ iteration, as determined using the extended exploitation tactic (first tactic). $X_{L-exploit}^{t+1}$ is Aquila's next updated position at the next $t+1^{th}$ iteration, as generated by the limited exploitation tactic (second tactic). t means the present iteration, and T is the maximum number of iterations.

The pseudo-code defining the original AO algorithm is offered in Algorithm 1 after presenting the two stages of the AO algorithm outlined above, the exploration and exploitation stages, and displaying the four tactics that Aquila relies on to catch the prey.

Algorithm 1 The original AO algorithm**Input:**

N – overall number of candidate positions (size of population)
 T – maximum number of allowed iterations
 D – problem's dimension size
 LB – lower limits of search space
 UB – upper limits of search space

Output:

X_{Best} – the comprehensive best position discovered while searching
 $f(X_{Best})$ – the comprehensive best fitness discovered, which should be decreased

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1: Start
2: Initialize a population of  $N$  positions, and set the pertinent parameters' values ( $s, \beta, r_1, U, \omega, \alpha, \delta$ );
3: Assign a random position  $X_i^0$  in the initial population;
4: Evaluate the fitness values  $f(X)$  for each position in the initial population;
5: Arrange the positions ascendingly based on their fitness function values  $f(X)$ ;
6: Locate the current best position  $X_{Best}^0$  of the lowest fitness, and its fitness value  $f(X_{Best}^0)$  through all positions in the initial population;
7:  $t \leftarrow 1$ ; ▷ present iteration number
8: while  $t < T$  do
9: Using the predefined values of the pertinent parameters, calculate  $y$  and  $z$ , according to Eq. (5) and (6);
10: Using the value of  $t$  and  $T$ , compute the value of  $QF^t$ , based on Eq. (10);
11: Obtain the value of  $G_1$  and  $G_2$ , utilizing Eq. (11);
12: for position  $i = 1 : N$  do
13: Assign the mean value of the current position  $X_i^t$  based on Eq. (2);
14: Update the levy flight distribution function  $Levy(D)$  using Eq. (4);
15: if  $t \leq (\frac{2}{3}) * T$  then
16: if  $rand < 0.5$  then
17: upgrade the current position through Eq. (1) to obtain  $X_{E-expl}^{t+1}$  by extended exploration tactic;
18:  $X_i^{t+1} \leftarrow X_{E-expl}^{t+1}$ ;
19: else if  $rand \geq 0.5$  then
20: upgrade the current position based on Eq. (3) to get  $X_{L-expl}^{t+1}$  by limited exploration tactic;
21:  $X_i^{t+1} \leftarrow X_{L-expl}^{t+1}$ ;
22: end if
23: else if  $t > (\frac{2}{3}) * T$  then
24: if  $rand < 0.5$  then
25: update the current position through Eq. (8) to obtain  $X_{E-exploit}^{t+1}$  by extended exploitation tactic;
26:  $X_i^{t+1} \leftarrow X_{E-exploit}^{t+1}$ ;
27: else if  $rand \geq 0.5$  then
28: update the current position using Eq. (9) to get  $X_{L-exploit}^{t+1}$  by limited exploitation tactic;
29:  $X_i^{t+1} \leftarrow X_{L-exploit}^{t+1}$ ;
30: end if
31: end if
32:  $f(X_i^{t+1}) \leftarrow$  Assess the value of fitness function for  $X_i^{t+1}$ ;
33: if  $f(X_i^{t+1}) < f(X_i^t)$  then ▷ Update the previous position if the present novel position is better than it
34:  $X_i^t \leftarrow X_i^{t+1}$ ;  $f(X_i^t) \leftarrow f(X_i^{t+1})$ ;
35: end if
36: end for
37: Re-arrange the positions ascendingly based on their fitness function values  $f(X)$ ;
38: Locate the best position  $X_{Best}^{t+1}$  and its best fitness value  $f(X_{Best}^{t+1})$  at completion of the present iteration  $t + 1$ ;
39:  $X_{Best} \leftarrow X_{Best}^{t+1}$ ;  $f(X_{Best}) \leftarrow f(X_{Best}^{t+1})$ ;
40:  $t \leftarrow t + 1$ ;
41: end while
42: End

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2.2. Presented algorithms for classification**2.2.1. k -Nearest neighbor (k -NN) algorithm**

One of the most popular pattern classification algorithms is k -NN (Alpaydin, 2020; Cunningham & Delany, 2021). The main aim of this algorithm is to forecast the classifying of a new sample's instances to a specific class by defining the closest class to the new instances with the majority of votes. For datasets that will be used with more complicated algorithms, k -NN is a good tool for assessing them.

The k -NN partitions the dataset into two sets: training and testing. Classification rules are created using the training set. In terms of the testing set, k -NN calculates the distances between the novel un-labeled testing sample and its closest k training neighbors for locating the training's k set's instances that are closest to the testing set's instances, k is usually a positive integer tiny value parameter ($k = 5$ Abd El-Mageed

et al., 2022; Thaher, Heidari, Mafarja, Dong, & Mirjalili, 2020). It is the only parameter that is considered for classifying novel testing sets. Eventually, the testing instance is classified using the most frequent class shared by its neighbors.

The distance metric used to determine the instances in the training's k set that are closest to the instances in the testing set is the Euclidean distance metric, which is expressed as follows:

$$ED = \sqrt{\sum_{j=1}^D (Att_{train_j} - Att_{test_j})^2} \quad (13)$$

where ED represents the Euclidean distance metric. D is the size of the problem's dimensional space. Att_{train_j} is a sample's j th attribute of a training dataset. Att_{test_j} is a sample's j th attribute of a testing dataset.

2.2.2. Support Vector Machine (SVM) algorithm

Support Vector Machine (SVM) (Tharwat, Hassanien, & Elnaghi, 2017) is a machine learning algorithm that may be employed to discover the best maximum margin hyper-planes in the space for isolating and perfectly classifying training samples into distinct classes. Even with a few training samples, SVM can process high-dimensional data with a short training duration and fewer resources for computing. After a few iterations, it can also detect and delete corrupted and noisy bands, which can significantly impact classification precisions.

SVM uses a margin maximization approach to evaluate the distributions related to statistics of various classes in the hyper-dimensional feature space. It forms hyper-planes to generate resolution limitations, whether for linear classification or may be adapted to nonlinear classification. SVM employs the so-called kernel function (Schölkopf, Smola, Bach, et al., 2002) as a scalar product to make the data linearly separable in nonlinear classification, in which the classes cannot be partitioned with a straight line. The linear kernel, radial basis function kernel, and polynomial kernel of degree d are the most often used kernel functions in SVM.

3. Proposed IBAO algorithm for FS

FS (Venkatesh & Anuradha, 2019) is a dimensionality-lessening strategy that allows a classifier to learn more precisely from the least relevant and informative subset of features by discarding irrelevant and noisy features that have no negative impact on the classifier's efficacy. This paper suggests that the FS strategy is implemented using an adjusted version of the AO algorithm dubbed the IBAO algorithm, which stands for an Improved Binary AO algorithm. The binarization idea is used to distinguish between the relevant and irrelevant features. The pertinent features that will be chosen are set to one, and the irrelevant features that will not be selected are set to zero.

The proposed IBAO algorithm's ability to handle the FS strategy depends on several steps, including initialization, upgrading of position using the AO algorithm, amendment of position randomly, binarization, fitness appreciation, and optimization using the Local Search (LS) strategy. The subsections that follow outline these steps.

3.1. Population initialization

The presented BAO starts with producing a random population of N potential positions. Each position indicates a feasible solution within its lower and upper limits constraints, depicted by the D dimensions' vector equivalent to the features' number in the primary benchmark. This generated random position is restricted within the $[-1, 1]$ range at each position vector variable.

3.2. Position upgrading through AO algorithm

Equations related to the AO algorithm, provided in Section 2.1, are employed to upgrade positions. These equations are then iterated until a specified halting condition is satisfied. The position vector is discretized after every upgrade to estimate the new position's fitness value using the classification error while preserving its premier form for a subsequent position update.

The maximum number of iterations T will be considered in this paper as the acceptable halting condition for properly evaluating the proposed algorithm's quality.

3.3. Random position amendment

The position amendment approach is preferred for repositioning and repairing the infeasible productive decision variables that vio-

late the search scope during position upgrading using meta-heuristic algorithms, such as the AO algorithm. For the values of decision variables that lie outside the bounds, the random position amendment approach is recommended for amending them to random values within the bounds. This method enhances the AO algorithm's exploitation procedure for randomly altering the optimal position. It is represented mathematically in the following equation:

$$X_{i,d}^{amend} = \begin{cases} X_{i,d}, & \text{if } X_d^{LB} \leq X_{i,d} \leq X_d^{UB} \\ rand(X_d^{LB}, X_d^{UB}), & \text{if } X_d^{LB} > X_{i,d} \text{ or } X_{i,d} > X_d^{UB} \end{cases} \quad (14)$$

where $X_{i,d}^{amend}$ is the valid value of the decision variable, $X_{i,d}$ is the value that exceeds the bounds of the variable, X_d^{LB} and X_d^{UB} represent the lower and upper limits for decision variable d , respectively, and $rand(X_d^{LB}, X_d^{UB})$ returns a random value between X_d^{LB} and X_d^{UB} with uniform distribution.

3.4. Binarization of position

Since the positions provided by the AO algorithm are continuous values, they cannot be employed directly to handle the binary FS strategy. It is necessary to take an intermediate step known as binarization to conform to the binary nature of FS, which turns the continuous values of positions into binary values. At the same time, the solution vector has the same length as the number of features in the original dataset.

Ones in the binary vector represent the continuous values of the relevant determinant features, whereas zeros represent the irrelevant undefined features' continuous values. At each iteration t , the mathematical formula for converting the continuous position X_i^t to a binary position $(X_i^t)_{bin}$ is as follows:

$$(X_i^t)_{bin} = \begin{cases} 1 & \text{if } X_i^t > thr_{rand}, \\ 0 & \text{otherwise.} \end{cases} \quad (15)$$

where $(X_i^t)_{bin}$ represents the resulted binary position obtained at t th iteration, X_i^t means the continuous position obtained at t th iteration, and thr_{rand} represents a random threshold value within the range $[0, 1]$.

3.5. Appreciation of fitness function

Two main opposing factors should be considered to estimate the quality of a solution: the accuracy and the size of selected features. The algorithm's prediction capability and quality improve when the accuracy of classifiers (k -NN and SVM classifiers) maximizes, whereas the chosen features' number in each solution minimizes. Nevertheless, the accuracy may be harmed if the size of chosen features is minimized more than intended. Hence there is a necessity for equilibrium between these two main factors. Accordingly, the fitness function for assessing the AO population will take care of striking an equilibrium among the chosen features' number (minimum) and the accuracy of (k -NN and SVM) classifiers, which can be mathematically depicted as follows:

$$Fit = w_1 \times (1 - accuracy) + w_2 \times \frac{|d^*|}{|D|}, \quad (16)$$

where $(1 - accuracy)$ means the classification error rate from (k -NN and SVM) classifiers, d^* indicates the size of the selected features, and D denotes the size of features in the original dataset. w_1 and w_2 represent weight parameters that address the relevance of accuracy and the selected features' length, respectively, where w_1 is an arbitrary number between $[0, 1]$, and $w_2 = 1 - w_1$. $w_1 = 0.99$ and $w_2 = 0.01$ based on extensive trials done in prior researches (Abdel-Basset, Ding, & El-Shahat, 2020; Faris et al., 2018). Instead of shortening the chosen features' length, the focus is on improving precision, which means that $w_1 > w_2$.

3.6. Exploration's quality optimization using LS strategy

This paper provides an improvement that combines the LS strategy with the BAO algorithm. It seeks to enhance the feature space's

Algorithm 2 LS strategy**Input:**

T_{LS} – maximum number of LS local iterations for applying LS strategy
 t – present iteration number
 X_{Best}^{t+1} – the best position obtained at the completion of BAO's present iteration $t + 1$

Output:

X_{Best} – the comprehensive best position discovered while searching
 $f(X_{Best})$ – the comprehensive best fitness discovered, which should be decreased

1: Start

```

2:  $X_{Temp}^{t+1} \leftarrow X_{Best}^{t+1}$ ; ▷ Temporary position used to store the value of  $X_{Best}^{t+1}$ 
3:  $t_{LS} \leftarrow 1$ ; ▷ Current LS iteration number for applying LS strategy
4: while  $t_{LS} < T_{LS}$  do
5:    $Indx_{LS\_feat} \leftarrow$  indices of four binary features, which selected randomly from  $X_{Temp}^{t+1}$ ;
6:   for  $indx_{each\_feat} \in Indx_{LS\_feat}$  do ▷  $indx_{each\_feat} \in \{0, 1\}$ , where 1 means a relevant selected feature and 0 means irrelevant deselected feature in  $X_{Temp}^{t+1}$ 
7:      $X_{Temp}^{t+1}[indx_{each\_feat}] \leftarrow \neg X_{Temp}^{t+1}[indx_{each\_feat}]$ ;
8:   end for
9:    $f(X_{Temp}^{t+1}) \leftarrow$  Assess the value of fitness function for  $X_{Temp}^{t+1}$ ;
10:  if  $f(X_{Temp}^{t+1}) < f(X_{Best}^{t+1})$  then
11:     $X_{Best}^{t+1} \leftarrow X_{Temp}^{t+1}$ ;  $f(X_{Best}^{t+1}) \leftarrow f(X_{Temp}^{t+1})$ ;
12:  end if
13:   $t_{LS} \leftarrow t_{LS} + 1$ ;
14: end while
15: End

```

exploration ability and the algorithm's performance by producing a new population with more proper positions while preserving its core structure.

LS strategy is applied after each present iteration $t + 1$ to improve the best position X_{Best}^{t+1} for the current iteration $t + 1$, knowing that the best position X_{Best}^{t+1} consists of a set of binary features. LS repeats many iterations T_{LS} . At each iteration of LS, four binary features are picked at random from the best position X_{Best}^{t+1} , then the binary value of each chosen feature is inverted, resulting in new offspring of the positions. Finally, in the manner illustrated in the preceding Section 3.5, the fitness value of these produced positions will be assessed and compared with the fitness value of the X_{Best}^{t+1} to determine whether the best position should be modified or left unchanged for the present iteration $t + 1$. Algorithm 2 displays the LS strategy's pseudo-code. It should be observed that it allows for avoiding the most deceptive positions and progressively improves the generated population's quality.

3.7. The overall IBAO algorithm

Algorithm 3 provides the pseudo-code for the steps of the IBAO algorithm proposed in the preceding subsections to handle the FS strategy. In addition, the proposed IBAO algorithm's flowchart demonstrating the IBAO algorithm's key steps is illustrated in Fig. 1. These steps include initializing the population and upgrading the position depending on the four AO's tactics mentioned earlier, which are mathematically divided into two stages: exploration and exploitation, random position adjustment, position binarization, fitness function evaluation, and optimization using the LS strategy.

3.8. Computational complexity of the IBAO algorithm

The overall proposed IBAO algorithm's computational complexity ($O(\text{IBAO})$) can be determined by computing the complexity of each step individually, which can be summarized in five essential steps, namely, population initialization, position upgrading, Random position amendment, appreciation of fitness function, and LS strategy. Hence:

$$\begin{aligned}
 O(\text{IBAO}) = & O(\text{Population initialization}) + O(\text{Position upgrading}) + \\
 & O(\text{Random position amendment}) \\
 & + O(\text{Appreciation of fitness function}) + \\
 & O(\text{LS strategy}).
 \end{aligned}
 \tag{17}$$

Assume that D represents the problem's dimensional space, N is the overall number of candidate positions (population size), and T means the maximum number of allowed iterations. The following can be acquired as follows:

$$O(\text{Population initialization}) = O(N).$$

$$O(\text{Position upgrading}) = O(N \times T \times D).$$

$$O(\text{Random position amendment}) = O(N \times T \times D).$$

$$O(\text{Appreciation of fitness function}) = O(N \times T).$$

$$O(\text{LS strategy}) = O(N \times T).$$

$$O(\text{Random position amendment} + \text{LS strategy}) = O(N \times T \times D) + O(N \times T) = O(N \times T \times D).$$

$$\text{Therefore, } O(\text{IBAO}) = O(N) + O(N \times T \times D) + O(N \times T \times D) + O(N \times T) = O(N \times T \times D).$$

4. Experimental results and analysis

This section describes the empirical results achieved by the suggested IBAO algorithm and their counterparts reported in the literature. The suggested algorithm is verified using 18 multi-scale benchmarks. Results are shown in terms of mean values for all evaluation metrics. The datasets used in all experiments to demonstrate the performance of the proposed algorithm are described in Section 4.1, main parameters of the meta-heuristic methods used in this paper are defined in Section 4.2, evaluation measures are explained in Section 4.3, results of the suggested IBAO in comparison with k -NN and SVM algorithms are investigated in Section 4.4, results of the suggested IBAO in comparison with their counterparts methods are investigated in Section 4.5, convergence graphs are plotted in Section 4.6, and finally, the Wilcoxon's test is presented in Section 4.7 to evaluate whether there are a reliable differences between the fitness rates obtained by the suggested IBAO algorithm and its peers.

4.1. Description of the benchmarks

Eighteen benchmarks with multi-scale features and instances obtained from the UCI machine learning repository Frank (2010) are employed in the entire experiments to demonstrate the performance of the proposed algorithm. This should support the validation of the proposed algorithm because each benchmark has a different number of features and instances. The description and properties of the benchmarks used in this paper and obtained from the UCI machine learning repository are presented in Table 1, which demonstrates the

Algorithm 3 The proposed IBAO algorithm**Input:**

N – overall number of candidate positions (size of population)
 T – maximum number of allowed iterations
 T_{LS} – maximum number of LS local iterations for applying LS strategy
 D – problem's dimension size
 LB, UB – lower limits and upper limits of search space

Output:

X_{Best} – the comprehensive best position discovered while searching
 $f(X_{Best})$ – the comprehensive best fitness discovered, which should be decreased

```

1: Start
2: Initialize a population of  $N$  positions, and set the pertinent parameters' values ( $s, \beta, r_1, U, \omega, \alpha, \delta$ );
3: Assign a random position  $X_i^0$  in the initial population;
4: Evaluate the values of fitness function  $f(X)$  for each position in the initial population;
5: Arrange the positions ascendingly based on their fitness function values  $f(X)$ ;
6: Locate the current best position  $X_{Best}^0$  of the lowest fitness, and its fitness value  $f(X_{Best}^0)$  in the initial population;
7:  $t \leftarrow 1$ ; ▷ Present iteration number
8: while  $t < T$  do
9: Using the predefined values of the pertinent parameters, calculate  $y$  and  $z$ , according to Eq. (5) and (6);
10: Using the value of  $t$  and  $T$ , compute the value of  $QF^t$ , based on Eq. (10);
11: Obtain the value of  $G_1$  and  $G_2$ , utilizing Eq. (11);
12: for position  $i = 1 : N$  do
13: Assign the mean value of the current position  $X_{Av}^t$  based on Eq. (2);
14: Update the levy flight distribution function  $Levy(D)$  using Eq. (4);
15: if  $t \leq (\frac{2}{3}) * T$  then
16: if  $rand < 0.5$  then
17: upgrade the current position through Eq. (1) to obtain  $X_{E-expl}^{t+1}$  by extended exploration tactic;
18:  $X_i^{t+1} \leftarrow X_{E-expl}^{t+1}$ ;
19: else if  $rand \geq 0.5$  then
20: upgrade the current position based on Eq. (3) to get  $X_{L-expl}^{t+1}$  by limited exploration tactic;
21:  $X_i^{t+1} \leftarrow X_{L-expl}^{t+1}$ ;
22: end if
23: else if  $t > (\frac{2}{3}) * T$  then
24: if  $rand < 0.5$  then
25: update the current position through Eq. (8) to obtain  $X_{E-exploit}^{t+1}$  by extended exploitation tactic;
26:  $X_i^{t+1} \leftarrow X_{E-exploit}^{t+1}$ ;
27: else if  $rand \geq 0.5$  then
28: update the current position using Eq. (9) to get  $X_{L-exploit}^{t+1}$  by limited exploitation tactic;
29:  $X_i^{t+1} \leftarrow X_{L-exploit}^{t+1}$ ;
30: end if
31: end if
32:  $f(X_i^{t+1}) \leftarrow$  Assess the value of fitness function for  $X_i^{t+1}$ ;
33: if  $f(X_i^{t+1}) < f(X_i^t)$  then ▷ Update the previous position if the present novel position is better than it
34:  $X_i^t \leftarrow X_i^{t+1}$ ;  $f(X_i^t) \leftarrow f(X_i^{t+1})$ ;
35: end if
36: end for
37: Re-arrange the positions ascendingly based on their fitness function values  $f(X)$ ;
38: Locate the best position  $X_{Best}^{t+1}$  and its best fitness value  $f(X_{Best}^{t+1})$  at completion of the present iteration  $t + 1$ ;
39:  $X_{Best}^{t+1} \leftarrow$  Apply the proposed LS strategy to enhance  $X_{Best}^{t+1}$  according to Algorithm 2;
40:  $X_{Best}^{t+1} \leftarrow X_{Best}^{t+1}$ ;  $f(X_{Best}^{t+1}) \leftarrow f(X_{Best}^{t+1})$ ;
41:  $t \leftarrow t + 1$ ;
42: end while
43: End

```

scientific name of each benchmark, the number of classification labels per benchmark, the number of instances (rows), and the number of features (attributes).

4.2. Parameters configuration

The suggested IBAO algorithm has been compared with binary conversions of various meta-heuristic algorithms, which include the original BAO algorithm and other twelve recent algorithms, containing Binary Salp Swarm Algorithm (BSSA), Binary Artificial Bee Colony (BABC), Binary Bat Algorithm (BBA), Binary Particle Swarm Optimization (BPSO), Binary Whale Optimization Algorithm (BWOA), Binary Grey Wolf Optimization (BGWO), Binary Grasshopper Optimization Algorithm (BGOA), Binary Sailfish Optimizer (BSFO), Binary Bird Swarm Algorithm (BBSA), Binary Atom Search Optimization (BASO), Binary

Harris Hawks Optimization (BHHO), and Binary Henry Gas Solubility Optimization (BHGSO). The main parameters of the ML classifiers used in this paper are defined as follows, the Euclidean distance metric k in k -NN classifier is estimated to be 5 (using results obtained from previous papers Abd El-Mageed et al., 2022; Mafarja & Mirjalili, 2018; Thaher et al., 2020). In contrast, the SVM classifier is built on a degree of 2 of the polynomial kernel. Thirty separate experiments are performed for each method in each dataset due to the meta-heuristic algorithms' stochastic nature. The different performance measures are reported in terms of mean results. For the comparison to be fair, the population size and the maximum number of iterations were set respectively to be 10 and 100 for all methods. In addition, the number of attributes in the datasets used here shows the size of the problem. At the same time, the continuous search domain is set to $[-1, 1]$ to

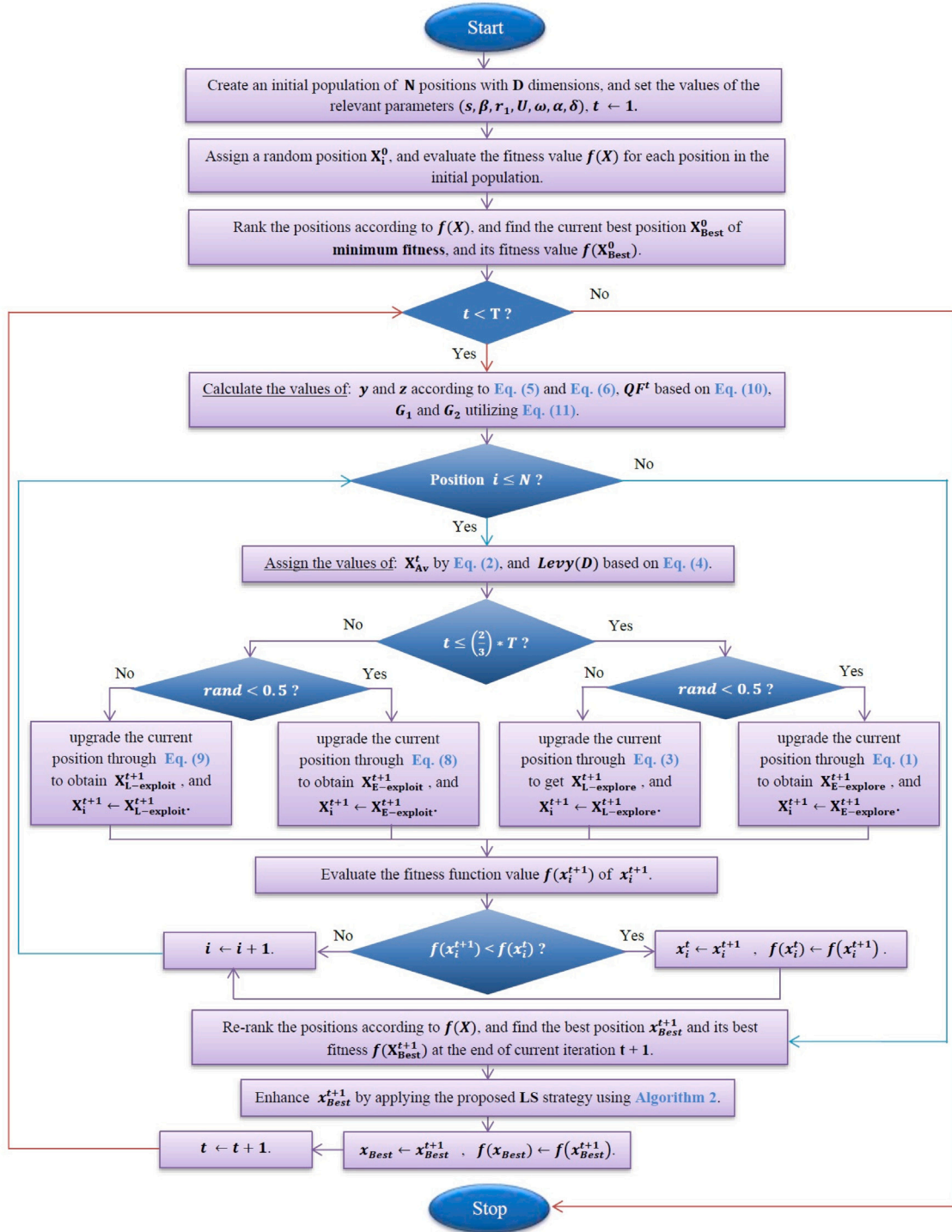


Fig. 1. Flowchart of the proposed IBAO algorithm.

allow individuals to search relatively widely but constrainedly within the continuous search space.

In the presented framework, the optimality degrees of the outcomes are confirmed using a ten-fold cross-validation method to assure the reliability of the values received where training and testing groups are distributed by splitting randomly every benchmark into two separate subsets, with training executed on 80% of the benchmark and testing on the rest (Faris et al., 2018). The training subset is utilized to learn the ML classifier through optimization, whereas the test subset is used to assess the chosen features. The remaining parameters are set for each method considering the original variants and the data presented in their

first publications. Standard configurations for all techniques, along with parameter settings for each method, are shown in Table 2. To execute the runs, Python is utilized in the computing environment with an Intel processor core i7, 16 GB of RAM, and an NVIDIA GTX 1050i GPU.

4.3. Metrics of performance

To appreciate the results of the suggested IBAO algorithm against its competitors, all methods are separately evaluated over 30 runs for each benchmark. To achieve this, the upcoming assessment measures for the FS strategy are used.

Table 1
Characteristics of the benchmarks utilized in this paper.

#	Benchmarks	No. of classes	No. of rows	No. of attributes
1	BreastEW	2	569	30
2	BreastCancer	2	699	9
3	Exactly	2	1000	13
4	Exactly2	2	1000	13
5	CongressEW	2	435	16
6	IonosphereEW	2	351	34
7	KrvskpEW	2	3196	36
8	HeartEW	2	270	13
9	M-of-n	2	1000	13
10	PenglungEW	7	73	325
11	Lymphography	4	148	18
12	SpectEW	2	267	22
13	SonarEW	2	208	60
14	Vote	2	300	16
15	Tic-tac-toe	2	958	9
16	WineEW	3	178	13
17	WaveformEW	3	5000	40
18	Zoo	7	101	16

- **Mean accuracy** (Mean_{Acc}): this measure is the correct data classification rate and is got by executing the method separately 30 runs:

$$\text{Mean}_{Acc} = \frac{1}{30} \frac{1}{m} \sum_{k=1}^{30} \sum_{r=1}^m \text{match}(PL_r, AL_r), \quad (18)$$

where Mean_{Acc} represents the mean accuracy, m is the volume of samples in the test subset, PL_r and AL_r respectively denote the output label of the predicted class and the reference class label for sample r , while $\text{match}(PL_r, AL_r)$ denotes a discriminant comparison function. If $PL_r == AL_r$, then $\text{match}(PL_r, AL_r) = 1$; otherwise, $\text{match}(PL_r, AL_r) = 0$.

- **Mean fitness value** (Mean_{Fit}): this measure calculates the mean fitness results achieved by running the suggested method separately for 30 runs, demonstrating the synergy between minimizing the number of features selected and reducing the error classification rate as Eq. (16). The minimum value presents the best result which is assessed according to fitness as:

$$\text{Mean}_{Fit} = \frac{1}{30} \sum_{k=1}^{30} f_*^k, \quad (19)$$

Table 2
Configurations of parameter for all algorithms.

Algorithm	Parameter
All algorithms	Run's number = 30 Iterations number $T = 100$ Population size $N = 10$ Dimensionality $D =$ The number of attributes in the used benchmarks
IBAO (Abualigah et al., 2021)	Scale factor whose value is defined by the scales of the problem $s = 0.01$ $\beta = 1.5$ in Eq. (4) Number of search cycles $r_1 = 10$ $U = 0.00565$ in Eqs. (5) and (6) $\omega = 0.005$ in Eqs. (5) and (6) Adjustment parameters for exploitation stage $\alpha = 0.1$ and $\delta = 0.1$ Aquila's random movements $G_1 \in [-1, 1]$ Flying slope of Aquila $G_2 \in [2, 0]$ LS local Iterations number $T_{LS} = 20$
BSSA	Number of scroungers $SD = 0.1 * N$ Number of producers $PD = 0.2 * N$ Maximum number of iterations in LSA = 20
BABC	Safety threshold $ST = 0.8$ Number of employed bees = 16 Number of scout bees = 3 Number of onlooker bees = 4
BPSO	Inertia weight ($\omega_{\max} = 0.9\omega_{\min} = 0.4$) Acceleration coefficients ($c_2 = c_1 = 1.2$)
BBA	Loudness $A = 0.8$ Lower and upper pulse frequencies = 0, 10 Pulse emission rate $r = 0.95$
BWOA	a is linearly reduced from 2 to 0 $b = 1.0$ $p = 0.5$
BSFO	Ratio between sardines and sailfish $pp = 0.1$ $\epsilon = 0.0001$ $A = 1$
BHHO	Rabbit energy $E \in [-1, 1]$
BGWO	a is linearly reduced from 2 to 0
BGOA	$C_{\min} = 0.00004$ and $C_{\max} = 1$
BBSA	Frequency of flight $ff = 10$ Followed coefficient $fl = 0.5$ Effect on birds' vigilance behaviors ($a_1 = a_2 = 1.0$) Acceleration coefficients ($c_1 = c_2 = 1.5$) Probability of foraging for food $p = 0.8$
BASO	Multiplier weight $\beta = 0.2$ Depth weight $\alpha = 50$
BHGSO	Number of clusters = 2 $l_1 = 5E - 03$, $l_2 = 1E + 02$, and $l_3 = 1E - 02$ $\alpha = \beta = 0.1$ and $K = 1.0$

Table 3

Performance comparisons of the basic k -NN algorithm and IBAO- k -NN according to the mean accuracy (Mean_{Acc}) and the mean number of chosen features (Mean_{Feat}).

Benchmarks	Accuracy			Feature		
	k -NN	IBAO- k -NN (Mean)	Rate of increase (%)	k -NN	IBAO- k -NN (Mean)	Rate of decrease (%)
BreastEW	0.9211	0.9649	04.76%	030.00	004.73	84.23%
BreastCancer	0.6214	0.9857	58.63%	009.00	006.07	32.56%
Exactly	0.7350	1.0000	36.05%	013.00	006.00	53.85%
Exactly2	0.7400	0.7730	04.46%	013.00	006.30	51.54%
CongressEW	0.9310	0.9770	04.94%	016.00	005.43	66.06%
IonosphereEW	0.8451	0.9362	10.78%	034.00	008.93	73.74%
KrVsKpEW	0.9656	0.9791	01.40%	036.00	021.33	40.75%
HeartEW	0.6296	0.9074	44.12%	013.00	004.73	63.62%
M-of-n	0.8800	1.0000	13.64%	013.00	006.00	53.85%
PenglungEW	0.5333	0.6533	22.50%	325.00	108.97	66.47%
Lymphography	0.7000	0.8444	20.63%	018.00	007.23	59.83%
SpectEW	0.7778	0.8920	14.68%	022.00	007.27	66.95%
SonarEW	0.8571	0.9897	15.47%	060.00	020.87	65.22%
Vote	0.9333	1.0000	07.14%	016.00	003.20	80.00%
Tic-tac-toe	0.8441	0.8542	01.20%	009.00	009.00	00.00%
WineEW	0.5833	1.0000	71.44%	013.00	004.03	69.00%
WaveformEW	0.8040	0.8501	05.73%	040.00	022.40	44.00%
Zoo	0.9048	1.0000	10.52%	016.00	005.17	67.69%
Score (W T L)	0 0 18	18 0 0	*	0 1 17	17 1 0	*

where $Mean_{Fit}$ represents the mean fitness value, f_*^k is the optimum fitness result achieved in the k th run.

- **Mean number of features selected (Mean_{Feat}):** this measure shows the mean number of selected features by running the method separately 30 runs and is defined as:

$$Mean_{Feat} = \frac{1}{30} \sum_{k=1}^{30} \frac{|d_*^k|}{|D|}, \quad (20)$$

where $Mean_{Feat}$ represents the mean number of features selected, $|d_*^k|$ is the size of features chosen in the optimal solution for the k th run, and $|D|$ represents the complete size of features in the used benchmark.

- **Standard Deviation (SD):** using the previous outcomes, the mean results achieved from the 30 implementations of each method on every benchmark are assessed according to stability as:

$$SD = \sqrt{\frac{1}{29} \sum_{k=1}^{30} (Y_*^k - Mean_Y)^2}, \quad (21)$$

where SD represents the standard deviation, Y represents the measure to be evaluated, Y_*^k shows the measure value Y in the k th run, and $Mean_Y$ is the mean of the measure over 30 separate times.

The outcomes shown in the next tables are the mean values over 30 separate times according to the fitness value (Mean_{Fit}), accuracy (Mean_{Acc}), and the size of features selected (Mean_{Feat}). The empirical outcomes are discussed and analyzed closely in the following subsections, where **bold** values show the best results.

4.4. Experimental results of two machine learning classifiers (k -NN and SVM) and the proposed IBAO

The empirical results in this subsection show performance comparisons of the two presented machine learning classifiers (k -NN and SVM) and the suggested methods (IBAO- k -NN and IBAO-SVM) according to the mean accuracy (Mean_{Acc}) and the mean number of features selected (Mean_{Feat}), to assess the breadth and effect of the suggested IBAO.

4.4.1. Comparisons of k -NN and IBAO- k -NN

Comparisons of performance metrics between the suggested IBAO- k -NN and the basic k -NN algorithm based on the mean accuracy

of classification (Mean_{Acc}) and the mean number of chosen features (Mean_{Feat}) are presented in Table 3. By analyzing Table 3, it should be noted that Mean_{Acc} with the IBAO- k -NN method is increased on all benchmarks with the progressive rate of over 20% on 6 of them, while Mean_{Acc} itself had a score of more than 94% on 10 out of 18 benchmarks, achieving 100% mean accuracy for 5 of them. It should also be noted that the Mean_{Feat} has been reduced according to the suggested IBAO- k -NN method on 94.4% of the benchmarks. But, IBAO- k -NN could not enhance Mean_{Feat} on the Tic-tac-toe benchmark. Finally, the suggested IBAO- k -NN method exceeded the original k -NN algorithm overall benchmarks in terms of Mean_{Acc} and on the majority of benchmarks in terms of Mean_{Feat}. Therefore, the suggested IBAO- k -NN has presented promising results to the strategy of FS by comparing with the basic k -NN algorithm on the selected benchmarks.

4.4.2. Comparisons of SVM and IBAO-SVM

Comparisons of performance metrics between the suggested IBAO-SVM and the basic SVM algorithm based on the mean accuracy of classification (Mean_{Acc}) and the mean number of chosen features (Mean_{Feat}) are shown in Table 4. By analyzing Table 4, it should be noted that Mean_{Acc} was improved over 16 benchmarks with a reliable improvement of more than 44% on 2 of them, while Mean_{Acc} was obtained more than 90% on 12 benchmarks, achieving 100% mean accuracy for 4 of them. It should also be noted that the Mean_{Feat} has been reduced in favor of the suggested IBAO-SVM method on all benchmarks. But, IBAO-SVM was not able to enhance the mean Mean_{Acc} on the Exactly2, and M-of-n benchmarks. Finally, the suggested IBAO-SVM method exceeded the basic SVM method on 16 benchmarks in terms of Mean_{Acc} and on all benchmarks in terms of Mean_{Feat}. Therefore, the suggested IBAO-SVM has presented promising results to the task of FS by comparing with the basic SVM algorithm on the selected benchmarks.

4.5. Comparisons results of the proposed IBAO versus other state-of-art meta-heuristic algorithms

To demonstrate the dominance of the suggested IBAO algorithm over its counterparts using k -NN and SVM classifiers (i.e., the IBAO- k -NN and IBAO-SVM methods), IBAO was compared with other recent meta-heuristic methods, such as BABC, BSSA, BBA, BPSO, BWOA, BGWO, BSFO, BGOA, BHHO, BASO, BBSA, and BHGSO which are conducted in identical situations. Results are represented in terms of the mean fitness value, mean accuracy, and mean number of features selected.

Table 4

Performance comparisons of the basic SVM algorithm and IBAO-SVM according to mean accuracy (Mean_{Acc}) and the mean number of chosen features (Mean_{F_{cal}}).

Benchmarks	Accuracy			Feature		
	SVM	IBAO-SVM (Mean)	Rate of increase (%)	SVM	IBAO-SVM (Mean)	Rate of decrease (%)
BreastEW	0.8947	0.9474	05.89%	030.00	005.77	80.76%
BreastCancer	0.6786	0.9786	44.21%	009.00	005.10	43.33%
Exactly	0.6860	0.7450	08.60%	013.00	006.30	51.54%
Exactly2	0.7500	0.7500	00.00%	013.00	001.00	92.31%
CongressEW	0.9655	0.9770	01.19%	016.00	003.47	78.31%
IonosphereEW	0.8873	0.9653	08.79%	034.00	011.33	66.68%
KrVsKpEW	0.9766	0.9852	00.88%	036.00	022.50	37.50%
HeartEW	0.7963	0.9074	13.95%	013.00	004.07	68.69%
M-of-n	1.0000	1.0000	00.00%	013.00	006.07	53.31%
PenglungEW	0.8267	0.8356	01.08%	325.00	102.83	68.36%
Lymphography	0.7667	0.8556	11.60%	018.00	007.87	56.28%
SpectEW	0.7407	0.8327	12.42%	022.00	007.60	65.45%
SonarEW	0.8333	0.9532	14.39%	060.00	022.10	63.17%
Vote	0.9333	1.0000	07.15%	016.00	003.40	78.75%
Tic-tac-toe	0.9049	0.9062	00.14%	009.00	008.00	11.11%
WineEW	0.6667	1.0000	49.99%	013.00	002.17	83.31%
WaveformEW	0.8650	0.8801	01.75%	040.00	023.20	42.00%
Zoo	0.8571	1.0000	16.67%	016.00	005.13	67.94%
Score (W T L)	0 2 16	16 2 0	*	0 0 18	18 0 0	*

Table 5

Comparisons of IBAO with a few meta-heuristic methods using k -NN classifier according to the mean fitness value (Mean_{F_{fit}}).

Benchmarks	Metric	BAO	IBAO	BSSA	BABC	BPSO	BBA	BGWO	BWOA	BGOA	BSFO	BHHO	BBSA	BASO	BHGSO
BreastEW	Mean	0.0367	0.0363	0.0372	0.0379	0.0394	0.0428	0.0375	0.0388	0.0381	0.0374	0.0388	0.0389	0.0473	0.0462
	SD	0.0007	0.0003	0.0007	0.0006	0.0027	0.0045	0.0005	0.0021	0.0007	0.0003	0.0029	0.0026	0.0066	0.0038
BreastCancer	Mean	0.0203	0.0202	0.0208	0.0203	0.0227	0.0246	0.0206	0.0210	0.0211	0.0201	0.0215	0.0208	0.0304	0.0232
	SD	0.0006	0.0004	0.0010	0.0005	0.0025	0.0037	0.0009	0.0016	0.0016	0.0000	0.0021	0.0015	0.0050	0.0022
Exactly	Mean	0.0056	0.0046	0.0177	0.0132	0.1232	0.1546	0.0260	0.0472	0.0390	0.0147	0.0302	0.0439	0.2710	0.1476
	SD	0.0055	0.0000	0.0213	0.0097	0.0764	0.0915	0.0354	0.0518	0.0342	0.0178	0.0374	0.0480	0.0535	0.0796
Exactly2	Mean	0.2325	0.2296	0.2331	0.2294	0.2332	0.2418	0.2324	0.2331	0.2324	0.2274	0.2369	0.2328	0.2450	0.2438
	SD	0.0040	0.0024	0.0044	0.0023	0.0049	0.0054	0.0041	0.0049	0.0037	0.0000	0.0053	0.0038	0.0063	0.0063
CongressEW	Mean	0.0283	0.0262	0.0281	0.0274	0.0349	0.0350	0.0281	0.0313	0.0296	0.0260	0.0312	0.0277	0.0382	0.0380
	SD	0.0035	0.0005	0.0036	0.0024	0.0039	0.0041	0.0038	0.0048	0.0042	0.0005	0.0046	0.0029	0.0038	0.0007
IonosphereEW	Mean	0.0665	0.0658	0.0792	0.0832	0.0970	0.1086	0.0806	0.0879	0.0878	0.0767	0.0788	0.0838	0.1100	0.1136
	SD	0.0128	0.0092	0.0112	0.0078	0.0109	0.0131	0.0141	0.0092	0.0099	0.0081	0.0102	0.0099	0.0083	0.0074
KrVsKpEW	Mean	0.0248	0.0234	0.0278	0.0266	0.0389	0.0421	0.0295	0.0336	0.0326	0.0292	0.0296	0.0325	0.0614	0.0384
	SD	0.0027	0.0025	0.0036	0.0024	0.0059	0.0097	0.0025	0.0049	0.0033	0.0025	0.0045	0.0051	0.0214	0.0044
HeartEW	Mean	0.0955	0.0953	0.0960	0.0958	0.1098	0.1178	0.0964	0.1022	0.0989	0.0951	0.0981	0.0986	0.1576	0.1179
	SD	0.0005	0.0006	0.0009	0.0008	0.0151	0.0176	0.0033	0.0104	0.0062	0.0005	0.0052	0.0062	0.0405	0.0144
M-of-n	Mean	0.0046	0.0046	0.0060	0.0070	0.0484	0.0491	0.0049	0.0135	0.0124	0.0054	0.0080	0.0097	0.1338	0.0515
	SD	0.0000	0.0000	0.0041	0.0048	0.0334	0.0508	0.0010	0.0140	0.0127	0.0021	0.0066	0.0082	0.0408	0.0242
PenglungEW	Mean	0.3562	0.3466	0.3826	0.3742	0.3962	0.3985	0.3805	0.3939	0.3873	0.3586	0.3781	0.3895	0.4050	0.4012
	SD	0.0309	0.0312	0.0288	0.0321	0.0162	0.0116	0.0300	0.0196	0.0261	0.0312	0.0308	0.0242	0.0163	0.0002
Lymphography	Mean	0.1665	0.1580	0.1672	0.1682	0.1926	0.2238	0.1695	0.1784	0.1740	0.1646	0.1793	0.1746	0.2331	0.2145
	SD	0.0142	0.0150	0.0144	0.0060	0.0172	0.0227	0.0189	0.0167	0.0161	0.0107	0.0225	0.0119	0.0192	0.0193
SpectEW	Mean	0.1112	0.1102	0.1187	0.1199	0.1367	0.1560	0.1181	0.1288	0.1265	0.1138	0.1232	0.1236	0.1621	0.1515
	SD	0.0066	0.0061	0.0095	0.0093	0.0111	0.0130	0.0100	0.0130	0.0107	0.0076	0.0123	0.0102	0.0117	0.0126
SonarEW	Mean	0.0209	0.0137	0.0236	0.0243	0.0409	0.0571	0.0225	0.0342	0.0275	0.0167	0.0251	0.0274	0.0605	0.0534
	SD	0.0103	0.0114	0.0088	0.0084	0.0114	0.0166	0.0098	0.0118	0.0075	0.0114	0.0093	0.0113	0.0167	0.0111
Vote	Mean	0.0027	0.0020	0.0024	0.0029	0.0103	0.0178	0.0024	0.0038	0.0040	0.0023	0.0060	0.0039	0.0220	0.0188
	SD	0.0027	0.0003	0.0006	0.0006	0.0078	0.0067	0.0005	0.0042	0.0040	0.0003	0.0062	0.0042	0.0100	0.0048
Tic-tac-toe	Mean	0.1567	0.1544	0.1554	0.1544	0.1576	0.1611	0.1547	0.1547	0.1551	0.1544	0.1610	0.1554	0.1908	0.1613
	SD	0.0042	0.0000	0.0030	0.0000	0.0050	0.0099	0.0018	0.0018	0.0026	0.0000	0.0112	0.0031	0.0166	0.0103
WineEW	Mean	0.0031	0.0031	0.0032	0.0033	0.0075	0.0144	0.0034	0.0046	0.0035	0.0031	0.0037	0.0036	0.0325	0.0087
	SD	0.0001	0.0001	0.0003	0.0004	0.0091	0.0133	0.0004	0.0049	0.0004	0.0000	0.0008	0.0005	0.0153	0.0075
WaveformEW	Mean	0.1572	0.1540	0.1627	0.1624	0.1766	0.1794	0.1634	0.1667	0.1672	0.1626	0.1621	0.1656	0.1932	0.1786
	SD	0.0042	0.0041	0.0054	0.0040	0.0054	0.0067	0.0059	0.0062	0.0048	0.0043	0.0066	0.0068	0.0077	0.0044
Zoo	Mean	0.0034	0.0032	0.0036	0.0036	0.0042	0.0095	0.0036	0.0039	0.0039	0.0034	0.0040	0.0039	0.0112	0.0068
	SD	0.0003	0.0002	0.0005	0.0004	0.0006	0.0138	0.0004	0.0005	0.0005	0.0003	0.0006	0.0004	0.0154	0.0080
Score	W T L	0 2 16	10 3 5	0 0 18	0 1 17	0 0 18	0 0 18	0 0 18	0 0 18	0 0 18	4 2 12	0 0 18	0 0 18	0 0 18	0 0 18

4.5.1. Comparisons based on the k -NN classifier

Table 5 shows the experimental results of comparisons based on the averaged fitness values between the proposed IBAO- k -NN and other well-developed state-of-art meta-heuristic optimization algorithms in

addressing FS strategy. By analyzing Table 5, IBAO- k -NN outperformed the other methods in 10 benchmarks and tied over 4 benchmarks in the FS problem, thus providing a significantly better effect in 14 out of 18 benchmarks, representing 77% of all benchmarks. In addition, The

Table 6Comparisons of IBAO with a few meta-heuristic methods using k -NN classifier according to the mean accuracy (Mean_{Acc}).

Benchmarks	Metric	BAO	IBAO	BSSA	BABC	BPSO	BBA	BGWO	BWOA	BGOA	BSFO	BHHO	BBSA	BASO	BHGSO
BreastEW	Mean	0.9649	0.9649	0.9649	0.9649	0.9640	0.9608	0.9649	0.9643	0.9649	0.9649	0.9637	0.9640	0.9567	0.9585
	SD	0.0000	0.0000	0.0000	0.0000	0.0026	0.0044	0.0000	0.0022	0.0000	0.0000	0.0030	0.0026	0.0064	0.0039
BreastCancer	Mean	0.9857	0.9857	0.9857	0.9857	0.9836	0.9817	0.9857	0.9852	0.9852	0.9857	0.9848	0.9855	0.9757	0.9843
	SD	0.0000	0.0000	0.0000	0.0000	0.0033	0.0044	0.0000	0.0018	0.0018	0.0000	0.0024	0.0013	0.0057	0.0029
Exactly	Mean	0.9990	1.0000	0.9872	0.9917	0.8818	0.8502	0.9788	0.9577	0.9660	0.9902	0.9747	0.9610	0.7333	0.8575
	SD	0.0054	0.0000	0.0211	0.0094	0.0764	0.0919	0.0353	0.0518	0.0341	0.0176	0.0373	0.0480	0.0541	0.0795
Exactly2	Mean	0.7703	0.7730	0.7697	0.7737	0.7695	0.7607	0.7703	0.7698	0.7707	0.7750	0.7658	0.7695	0.7572	0.7590
	SD	0.0039	0.0028	0.0046	0.0022	0.0049	0.0057	0.0045	0.0051	0.0038	0.0000	0.0058	0.0039	0.0073	0.0066
CongressEW	Mean	0.9747	0.9770	0.9747	0.9762	0.9678	0.9678	0.9747	0.9713	0.9736	0.9770	0.9709	0.9759	0.9651	0.9655
	SD	0.0046	0.0000	0.0046	0.0029	0.0046	0.0046	0.0046	0.0057	0.0053	0.0000	0.0057	0.0034	0.0036	0.0000
IonosphereEW	Mean	0.9357	0.9362	0.9235	0.9197	0.9061	0.8944	0.9221	0.9150	0.9150	0.9258	0.9239	0.9192	0.8930	0.8901
	SD	0.0130	0.0094	0.0113	0.0083	0.0111	0.0135	0.0144	0.0093	0.0099	0.0081	0.0100	0.0102	0.0086	0.0076
KrVsKpEW	Mean	0.9811	0.9824	0.9778	0.9794	0.9665	0.9637	0.9764	0.9720	0.9731	0.9766	0.9759	0.9735	0.9444	0.9682
	SD	0.0027	0.0026	0.0037	0.0026	0.0060	0.0103	0.0026	0.0051	0.0034	0.0026	0.0044	0.0050	0.0220	0.0046
HeartEW	Mean	0.9074	0.9074	0.9074	0.9074	0.8938	0.8858	0.9068	0.9019	0.9049	0.9074	0.9056	0.9049	0.8457	0.8864
	SD	0.0000	0.0000	0.0000	0.0000	0.0158	0.0186	0.0033	0.0109	0.0063	0.0000	0.0056	0.0063	0.0415	0.0149
M-of-n	Mean	1.0000	1.0000	0.9988	0.9978	0.9577	0.9563	0.9998	0.9917	0.9928	0.9995	0.9970	0.9955	0.8717	0.9545
	SD	0.0000	0.0000	0.0040	0.0046	0.0330	0.0508	0.0009	0.0136	0.0124	0.0020	0.0064	0.0079	0.0419	0.0239
PenglungEW	Mean	0.6444	0.6533	0.6178	0.6267	0.6044	0.6022	0.6200	0.6067	0.6133	0.6422	0.6222	0.6111	0.5956	0.6000
	SD	0.0314	0.0317	0.0295	0.0327	0.0166	0.0120	0.0306	0.0200	0.0267	0.0321	0.0314	0.0248	0.0166	0.0000
Lymphography	Mean	0.8356	0.8444	0.8356	0.8344	0.8100	0.7789	0.8333	0.8244	0.8289	0.8378	0.8233	0.8278	0.7700	0.7889
	SD	0.0147	0.0157	0.0147	0.0060	0.0175	0.0235	0.0192	0.0171	0.0166	0.0113	0.0230	0.0124	0.0199	0.0199
SpectEW	Mean	0.8907	0.8920	0.8840	0.8833	0.8660	0.8469	0.8846	0.8741	0.8765	0.8889	0.8796	0.8790	0.8407	0.8525
	SD	0.0073	0.0069	0.0095	0.0097	0.0114	0.0135	0.0103	0.0130	0.0110	0.0083	0.0124	0.0104	0.0123	0.0131
SonarEW	Mean	0.9825	0.9897	0.9802	0.9802	0.9635	0.9468	0.9817	0.9698	0.9770	0.9873	0.9786	0.9770	0.9437	0.9516
	SD	0.0105	0.0118	0.0089	0.0089	0.0119	0.0170	0.0101	0.0122	0.0075	0.0119	0.0094	0.0115	0.0168	0.0115
Vote	Mean	0.9994	1.0000	1.0000	1.0000	0.9928	0.9850	1.0000	0.9989	0.9989	1.0000	0.9967	0.9989	0.9811	0.9850
	SD	0.0030	0.0000	0.0000	0.0000	0.0083	0.0066	0.0000	0.0042	0.0042	0.0000	0.0067	0.0042	0.0103	0.0050
Tic-tac-toe	Mean	0.8505	0.8542	0.8526	0.8542	0.8493	0.8453	0.8536	0.8536	0.8531	0.8542	0.8460	0.8526	0.8135	0.8453
	SD	0.0066	0.0000	0.0047	0.0000	0.0075	0.0119	0.0028	0.0028	0.0039	0.0000	0.0128	0.0047	0.0167	0.0121
WineEW	Mean	1.0000	1.0000	1.0000	1.0000	0.9963	0.9898	1.0000	0.9991	1.0000	1.0000	1.0000	1.0000	0.9722	0.9972
	SD	0.0000	0.0000	0.0000	0.0000	0.0094	0.0134	0.0000	0.0050	0.0000	0.0000	0.0000	0.0000	0.0160	0.0083
WaveformEW	Mean	0.8470	0.8501	0.8416	0.8424	0.8273	0.8249	0.8407	0.8373	0.8370	0.8417	0.8423	0.8386	0.8112	0.8259
	SD	0.0043	0.0043	0.0054	0.0040	0.0056	0.0071	0.0059	0.0063	0.0048	0.0044	0.0066	0.0069	0.0079	0.0045
Zoo	Mean	1.0000	1.0000	1.0000	1.0000	1.0000	0.9952	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9937	0.9984
	SD	0.0000	0.0000	0.0000	0.0000	0.0000	0.0143	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0162	0.0085
Score	W T L	0 6 12	8 9 1	0 6 12	0 7 11	0 1 17	0 0 18	0 5 13	0 1 17	0 3 15	1 8 9	0 2 16	0 2 16	0 0 18	0 0 18

benchmark utilized in this paper contains both large and small-scale benchmarks, proving the capability of IBAO- k -NN to perform consistently on entire benchmarks nevertheless of the size of the benchmark. For the 4 benchmarks missing by IBAO- k -NN, the results achieved by IBAO- k -NN are very close to other methods when comparing the mean fitness values. This reveals that the superior outcomes achieved by IBAO- k -NN. With the exception of BSFO, none of the competitor's methods compared to IBAO- k -NN are first in any of the benchmarks, which ensures the superiority of the suggested IBAO- k -NN. Also, the stability of the proposed IBAO- k -NN is relatively strong depend on the SD values of the different algorithms. Finally, it should be noted that IBAO- k -NN is a superior method, with a stable ability to avoid local optima and a balance between exploration and exploitation within the solution space through iterations. While, other methods may become trapped with the local optima problem.

Moreover, Table 6 presents the empirical outcomes of comparisons based on the mean accuracy values of classification between the presented IBAO- k -NN and other well-developed state-of-art meta-heuristic optimization algorithms in addressing FS strategy. From Table 6, it should be noted that the suggested IBAO- k -NN outperformed all other methods based on mean values of accuracy across eight benchmarks while achieving equal results with other methods across nine benchmarks, providing significantly better effect in 17 out of 18 benchmarks. In addition, consider that the BSFO method achieved second rank with a slight improvement on only one benchmark (Exactly2) by a margin of 0.0020% while tying on eight benchmarks.

Additionally, Table 7 exhibits the results of comparisons based on the mean number of features selected between the proposed IBAO- k -NN and other well-developed state-of-art meta-heuristic optimization algorithms in addressing FS strategy. as presented in Table 7, the proposed IBAO- k -NN obtained the exploration capability better than other algorithms in terms of the mean selected features number which was confirmed by choosing the least features number over 9 (8 wins and one tie), better than BSFO, BHHO, BASO, BHHO, BAO, and BASO, which chose fewer irrelevant features than the proposed IBAO- k -NN and other methods over only 3, 2, 2, and 2 benchmarks sequentially. This verifies the capability of the proposed IBAO- k -NN to neglect searching non-significant search regions and discover the most feasible regions. Therefore, the proposed IBAO- k -NN can minimize the feature search region by identifying the most relevant attributes while preserving the highest classification accuracy.

4.5.2. Comparisons based on the SVM classifier

The comparison results between the proposed IBAO-SVM and other well-developed state-of-art meta-heuristic optimizers are illustrated in Table 8 relied on the mean fitness values. By analyzing Table 8, IBAO-SVM outperformed the other methods in 13 benchmarks and tied over three benchmarks in the FS problem, thus having a significantly better effect in 16 out of 18 benchmarks, representing 88% of all benchmarks. In addition, The benchmark utilized in this paper contains both large and small-scale benchmarks, proving the capability of IBAO-SVM to perform consistently on entire benchmarks nevertheless of the size of the benchmarks. For the two benchmarks missing by IBAO-SVM,

Table 7Comparisons of IBAO with a few meta-heuristic methods using k -NN classifier according to the mean number of features selected (Mean_{Feat}).

Benchmarks	Metric	BAO	IBAO	BSSA	BABC	BPSO	BBA	BGWO	BWOA	BGOA	BSFO	BHHO	BBSA	BASO	BHGSO
BreastEW	Mean	005.90	004.73	007.53	009.43	011.27	012.00	008.43	010.57	010.23	007.97	008.73	009.87	013.43	015.23
	SD	002.02	000.89	002.09	001.73	001.75	002.03	001.56	001.98	002.06	001.05	002.06	001.63	002.91	001.45
BreastCancer	Mean	006.20	006.07	006.70	006.13	006.40	006.43	006.43	006.43	006.53	006.00	006.43	006.43	006.40	007.60
	SD	000.60	000.36	001.01	000.50	001.60	001.36	000.88	001.12	000.99	000.00	001.12	000.88	001.33	001.43
Exactly	Mean	006.03	006.00	006.47	006.47	008.10	008.20	006.57	006.90	007.00	006.43	006.67	006.93	009.13	008.50
	SD	000.18	000.00	000.56	000.50	001.04	001.28	000.67	000.79	000.63	000.56	000.70	000.73	001.82	001.18
Exactly2	Mean	006.63	006.30	006.57	006.97	006.47	006.37	006.53	006.77	007.00	006.00	006.53	006.00	005.97	006.83
	SD	001.38	000.97	001.23	001.02	001.02	001.74	001.09	001.23	001.39	000.00	001.61	001.15	002.23	001.85
CongressEW	Mean	005.17	005.43	004.90	006.27	004.83	005.03	004.93	004.53	005.43	005.17	003.83	006.07	005.97	006.10
	SD	002.27	000.84	001.85	001.53	001.83	001.64	001.77	001.84	002.17	000.73	002.08	001.59	001.58	001.11
IonosphereEW	Mean	009.70	008.93	011.80	012.57	013.70	013.80	011.57	012.97	012.57	011.13	011.93	013.07	013.73	016.53
	SD	002.45	002.13	003.05	002.63	002.95	002.93	002.54	001.87	002.93	002.81	002.41	002.89	003.44	003.11
KrVsKpEW	Mean	022.07	021.33	020.93	022.33	020.83	022.13	021.80	021.37	021.73	021.70	020.70	022.60	022.93	024.93
	SD	002.73	001.87	002.26	002.43	001.93	003.27	002.36	002.21	002.77	002.21	003.21	002.80	004.19	002.91
HeartEW	Mean	005.03	004.73	005.67	005.43	006.03	006.17	005.37	006.53	006.23	004.40	005.97	005.83	006.27	007.03
	SD	000.71	000.81	001.11	001.09	001.38	001.53	001.35	001.26	001.17	000.71	001.17	001.49	001.34	001.14
M-of-n	Mean	006.00	006.00	006.30	006.37	008.50	007.67	006.20	006.83	006.90	006.37	006.50	006.77	008.73	008.40
	SD	000.00	000.00	000.46	000.55	001.48	001.27	000.40	000.86	000.75	000.48	000.56	000.72	002.02	000.99
PenglungEW	Mean	137.37	108.97	137.10	150.27	148.23	151.43	139.27	146.40	147.80	144.37	131.80	146.80	148.30	168.87
	SD	010.54	011.75	018.46	011.97	012.03	017.68	011.15	012.66	014.16	022.08	015.37	015.29	024.21	007.92
Lymphography	Mean	006.67	007.23	007.93	007.67	008.13	008.77	008.10	008.37	008.27	007.27	007.90	007.43	009.77	009.97
	SD	001.33	001.28	001.69	001.60	001.77	001.91	001.54	001.30	001.36	001.29	001.66	001.61	001.89	001.58
SpectEW	Mean	006.70	007.27	008.40	009.63	009.00	009.83	008.43	009.03	009.40	008.27	008.87	008.47	009.77	012.03
	SD	002.19	002.32	002.30	002.56	001.26	002.21	002.09	001.87	002.09	002.13	002.26	001.50	002.47	002.20
SonarEW	Mean	021.93	020.87	023.63	027.67	028.33	026.83	026.30	026.20	028.13	024.87	023.57	027.77	028.27	033.10
	SD	003.36	004.35	004.16	004.21	003.52	004.57	003.36	003.30	004.09	004.13	003.72	003.48	005.64	003.99
Vote	Mean	003.43	003.20	003.83	004.67	005.07	004.77	003.80	004.27	004.60	003.67	004.30	004.50	005.33	006.33
	SD	000.84	000.40	001.00	000.94	001.29	001.33	000.83	001.09	000.99	000.47	001.37	000.96	001.37	001.08
Tic-tac-toe	Mean	007.83	009.00	008.50	009.00	007.53	007.13	008.83	008.83	008.70	009.00	007.70	008.53	005.60	007.33
	SD	002.11	000.00	001.50	000.00	002.25	002.16	000.90	000.90	001.13	000.00	001.93	001.41	001.28	002.12
WineEW	Mean	004.03	004.03	004.17	004.33	005.03	005.67	004.37	004.73	004.53	004.00	004.80	004.63	006.47	007.73
	SD	000.18	000.18	000.37	000.47	001.02	001.27	000.55	001.03	000.56	000.00	000.98	000.71	001.69	001.77
WaveformEW	Mean	023.03	022.40	023.53	025.30	022.70	024.17	022.73	022.60	023.40	023.63	023.93	023.37	025.10	025.03
	SD	003.05	002.64	002.62	003.84	002.70	003.95	003.05	002.87	002.63	002.70	002.72	002.68	004.46	002.54
Zoo	Mean	005.40	005.17	005.83	005.77	006.73	007.67	005.80	006.30	006.30	005.37	006.33	006.17	007.87	008.43
	SD	000.49	000.37	000.78	000.62	001.00	001.45	000.65	000.86	000.74	000.48	000.91	000.64	002.05	001.33
Score	W T L	2 1 15	8 1 9	0 0 18	0 0 18	0 0 18	0 0 18	0 0 18	0 0 18	0 0 18	3 0 15	2 0 16	0 0 18	2 0 16	0 0 18

the results achieved by IBAO-SVM are very close to other methods when comparing the mean fitness values. This reveals the superior outcomes achieved by IBAO-SVM. Except for BSFO and BAO, none of the competitor's methods compared to IBAO-k-NN were first in any of the benchmarks, ensuring the superiority of the suggested IBAO-SVM. Also, the stability of the proposed IBAO-SVM is relatively strong, depending on the SD values of the different algorithms. Finally, it should be noted that IBAO-SVM is a superior method, with a stable ability to avoid local optima and a balance between exploitation and exploration within the solution space through iterations. In contrast, other methods may become trapped with the local optima problem.

Moreover, Table 9 demonstrates the empirical outcomes of comparisons between the presented IBAO-SVM and other well-developed state-of-art meta-heuristic optimization algorithms according to the mean accuracy values of classification in addressing FS strategy. Table 9 shows that the suggested IBAO-SVM outperformed all other methods based on mean accuracy values across seven benchmarks while achieving comparable results with other methods across 11 benchmarks, providing a significantly better effect in all benchmarks. The proposed IBAO-SVM had accuracy results of 100% for the M-of-n, Vote, WineEW, and Zoo benchmarks and classification accuracy results ranging from 96% to smaller than 100% for the BreastCancer, CongressEW, IonosphereEW, and KrVsKpEW benchmarks.

Based on the mean number of features chosen, Table 10 shows the empirical outcomes achieved through comparisons between the proposed IBAO-SVM and other sophisticated state-of-art meta-heuristic optimization algorithms in addressing FS strategy. It should be noted

that the proposed IBAO-SVM comes first by selecting the minimum mean size of attributes across 10 out of 18 benchmarks, followed by the BAO method, which chooses the minimum mean size of attributes across 4 out of 18 benchmarks. Additionally, as presented in Table 10, the proposed IBAO-SVM obtained the best exploration capability than other algorithms in terms of the mean selected features number which was confirmed by choosing the least features number over 11 (10 wins and one tie), better than BAO, BASO, and BWOA which chosen fewer irrelevant features number than the proposed IBAO-SVM and other methods over only 4, 2, and 1 benchmark sequentially. This verifies the capability of the proposed IBAO-SVM to neglect searching non-significant search regions and discover the most feasible areas. Therefore, the proposed IBAO-SVM can minimize the feature search region by identifying the most relevant attributes while preserving the highest classification accuracy.

4.6. Analysis of convergence

This section reveals an asymptotic investigation of the two presented methods (IBAO-k-NN and IBAO-SVM) for handling the FS strategy utilizing 18 widely used benchmarks to verify their capabilities in convergence, as per Figs. 2 and 3, sequentially. These convergence graphs are presented in Figs. 2 and 3 indicate the convergence capability of the suggested methods against their peers that are all evaluated and executed under identical situations of the number of iterations and population size. From Fig. 2, it is obvious that the

Table 8Comparisons of IBAO with a few meta-heuristic methods using SVM classifier according to the mean fitness value (Mean_{FIT}).

Benchmarks	Metric	BAO	IBAO	BSSA	BABC	BPSO	BBA	BGWO	BWOA	BGOA	BSFO	BHHO	BBSA	BASO	BHGSO
BreastEW	Mean	0.0545	0.0540	0.0574	0.0606	0.0639	0.0711	0.0581	0.0624	0.0633	0.0561	0.0592	0.0607	0.0708	0.0654
	SD	0.0007	0.0005	0.0041	0.0041	0.0027	0.0092	0.0040	0.0034	0.0046	0.0020	0.0050	0.0043	0.0081	0.0007
BreastCancer	Mean	0.0279	0.0263	0.0272	0.0263	0.0314	0.0339	0.0271	0.0287	0.0274	0.0262	0.0291	0.0281	0.0382	0.0320
	SD	0.0027	0.0003	0.0020	0.0003	0.0032	0.0061	0.0020	0.0029	0.0020	0.0000	0.0030	0.0029	0.0037	0.0031
Exactly	Mean	0.2702	0.2573	0.2940	0.2612	0.3006	0.3248	0.2835	0.2969	0.2892	0.2589	0.3044	0.3012	0.3321	0.3160
	SD	0.0260	0.0004	0.0303	0.0137	0.0311	0.0135	0.0301	0.0305	0.0318	0.0084	0.0310	0.0289	0.0040	0.0236
Exactly2	Mean	0.2483	0.2483	0.2483	0.2486	0.2492	0.2494	0.2483	0.2487	0.2485	0.2483	0.2483	0.2487	0.2499	0.2497
	SD	0.0000	0.0000	0.0001	0.0004	0.0005	0.0006	0.0001	0.0005	0.0004	0.0000	0.0000	0.0004	0.0006	0.0006
CongressEW	Mean	0.0252	0.0249	0.0256	0.0261	0.0284	0.0318	0.0256	0.0270	0.0266	0.0254	0.0259	0.0262	0.0359	0.0301
	SD	0.0005	0.0004	0.0006	0.0006	0.0029	0.0046	0.0006	0.0021	0.0020	0.0004	0.0007	0.0022	0.0046	0.0037
IonosphereEW	Mean	0.0378	0.0377	0.0438	0.0451	0.0589	0.0671	0.0441	0.0504	0.0525	0.0412	0.0457	0.0533	0.0783	0.0641
	SD	0.0076	0.0076	0.0078	0.0074	0.0082	0.0121	0.0067	0.0080	0.0075	0.0064	0.0094	0.0067	0.0098	0.0076
KrVsKpEW	Mean	0.0234	0.0209	0.0270	0.0266	0.0395	0.0402	0.0256	0.0315	0.0324	0.0284	0.0278	0.0304	0.0539	0.0362
	SD	0.0028	0.0031	0.0042	0.0031	0.0083	0.0083	0.0040	0.0048	0.0037	0.0023	0.0050	0.0031	0.0166	0.0037
HeartEW	Mean	0.0956	0.0948	0.0970	0.0980	0.1187	0.1405	0.1053	0.1109	0.1108	0.0952	0.1084	0.1100	0.1669	0.1465
	SD	0.0033	0.0002	0.0055	0.0061	0.0169	0.0240	0.0123	0.0134	0.0138	0.0004	0.0136	0.0168	0.0340	0.0224
M-of-n	Mean	0.0046	0.0047	0.0050	0.0053	0.0266	0.0419	0.0050	0.0102	0.0056	0.0051	0.0052	0.0056	0.1149	0.0116
	SD	0.0000	0.0002	0.0005	0.0004	0.0511	0.0642	0.0005	0.0251	0.0004	0.0004	0.0006	0.0006	0.0701	0.0269
PenglungEW	Mean	0.1909	0.1660	0.1953	0.2004	0.2027	0.2027	0.1933	0.1980	0.1980	0.1889	0.1887	0.2002	0.2138	0.2032
	SD	0.0244	0.0328	0.0197	0.0118	0.0002	0.0170	0.0224	0.0164	0.0164	0.0260	0.0263	0.0118	0.0243	0.0001
Lymphography	Mean	0.1615	0.1474	0.1607	0.1633	0.1852	0.1975	0.1654	0.1773	0.1726	0.1518	0.1740	0.1699	0.2188	0.1924
	SD	0.0140	0.0153	0.0190	0.0131	0.0178	0.0202	0.0140	0.0182	0.0141	0.0162	0.0182	0.0118	0.0218	0.0151
SpectEW	Mean	0.1837	0.1691	0.1985	0.1854	0.2233	0.2320	0.1977	0.2072	0.1981	0.1754	0.2009	0.1962	0.2499	0.2382
	SD	0.0237	0.0099	0.0247	0.0186	0.0244	0.0259	0.0267	0.0281	0.0230	0.0097	0.0300	0.0232	0.0075	0.0173
SonarEW	Mean	0.0629	0.0500	0.0718	0.0828	0.1132	0.1291	0.0784	0.0925	0.0951	0.0737	0.0781	0.0887	0.1362	0.1216
	SD	0.0119	0.0142	0.0131	0.0150	0.0198	0.0207	0.0132	0.0173	0.0160	0.0102	0.0158	0.0168	0.0218	0.0177
Vote	Mean	0.0024	0.0021	0.0025	0.0033	0.0090	0.0136	0.0030	0.0042	0.0043	0.0026	0.0032	0.0033	0.0243	0.0107
	SD	0.0005	0.0004	0.0007	0.0008	0.0072	0.0101	0.0008	0.0032	0.0029	0.0006	0.0008	0.0010	0.0115	0.0076
Tic-tac-toe	Mean	0.1017	0.1017	0.1017	0.1017	0.1206	0.1197	0.1017	0.1051	0.1018	0.1017	0.1018	0.1017	0.1602	0.1129
	SD	0.0000	0.0000	0.0000	0.0000	0.0191	0.0251	0.0000	0.0090	0.0004	0.0000	0.0004	0.0002	0.0320	0.0120
WineEW	Mean	0.0017	0.0017	0.0022	0.0027	0.0174	0.0205	0.0033	0.0075	0.0060	0.0019	0.0026	0.0038	0.0356	0.0189
	SD	0.0005	0.0003	0.0009	0.0007	0.0138	0.0135	0.0050	0.0104	0.0081	0.0004	0.0009	0.0051	0.0201	0.0131
WaveformEW	Mean	0.1268	0.1245	0.1295	0.1315	0.1468	0.1435	0.1328	0.1357	0.1380	0.1333	0.1320	0.1352	0.1536	0.1412
	SD	0.0022	0.0029	0.0041	0.0029	0.0051	0.0076	0.0033	0.0045	0.0050	0.0028	0.0044	0.0046	0.0070	0.0048
Zoo	Mean	0.0033	0.0032	0.0036	0.0038	0.0064	0.0170	0.0038	0.0055	0.0043	0.0034	0.0052	0.0039	0.0361	0.0129
	SD	0.0003	0.0002	0.0007	0.0005	0.0082	0.0201	0.0006	0.0083	0.0008	0.0003	0.0083	0.0008	0.0212	0.0167
Score	W T L	1 3 14	13 3 2	0 2 16	0 1 17	0 0 18	0 0 18	0 2 16	0 0 18	0 0 18	1 2 15	0 1 17	0 1 17	0 0 18	0 0 18

suggested IBAO- k -NN method demonstrated fast yet optimal convergence behavior over seven benchmarks involving the BreastCancer, IonosphereEW, PenglungEW, SonarEW, SpectEW, WineEW, and WaveformEW. While the IBAO-SVM model wins on five benchmarks, including the KrVsKpEW, SonarEW, PenglungEW, SpectEW, and WaveformEW. Hence, the suggested algorithms (IBAO- k -NN and IBAO-SVM) additionally emphasize their ability to acquire the optimal solution on time, ensuring an effective balance between exploration and exploitation capabilities.

4.7. Wilcoxon's rank-sum test

Tables 11 and 12 depict the significance of analysis by performing the Wilcoxon test as a pair-wise test to assess whether there is a statistically considerable variation between the fitness results obtained through the suggested IBAO algorithm and its counterparts (Derrac, García, Molina, & Herrera, 2011). Wilcoxon test is a nonparametric test employed in a hypothesis testing situation, and it works as follows. Firstly, the difference among the outcomes of pair algorithms on j th of N problems, and afterward, these differences are ranked based on their absolute values. Secondly, the sum of positive ranks (R^+) and negative ranks (R^-) are calculated, and the minimum among them is recorded. If the recorded results return a significance level of less than 5%, then the rejection of the null hypothesis is supported and vice versa. According to the results presented in Tables 11 and 12, it is clear that the IBAO- k -NN and IBAO-SVM algorithms outperformed all other algorithms in all cases. Therefore all p-values that are listed in Tables 11 and 12 are less than 0.05 (5% significance level) that, indicate robust evidence

against the null hypothesis, and can reveal that the obtained results by the proposed method are statistically better and not happened by chance.

4.8. Comparison against different FS algorithms from the literature

Table 13 illustrates the experimental outcomes of comparisons based on the mean accuracy values of classification in handling FS strategy between the two suggested algorithms (i.e., IBAO- k -NN and IBAO-SVM) and other meta-heuristic optimization algorithms from the literature, such as the TCSSA3 (Aljarah et al., 2018), BGOA-EPD-Tour (Mafarja et al., 2018), GA (Emery, Zawbaa, & Hassanien, 2016), and BSSA_S3_CP (Faris et al., 2018).

By examining Table 13, it can be seen that the suggested IBAO-SVM exceeded the competing methods based on mean accuracy across six benchmarks, while IBAO- k -NN was rated in the second place for five benchmarks, followed by the BGOA-EPD-Tour, BSSA_S3_CP, and TCSSA3 for 4, 2, and 1 of 18 benchmarks. Thus, the suggested IBAO- k -NN and IBAO-SVM methods got more effective outcomes for dealing with the FS strategy than other methods in the literature by winning the first two places in the competition.

4.9. Discussion

Based on the empirical analysis, it can be demonstrated that the proposed IBAO with k -NN and SVM classifiers yielded more reliable results than other recent algorithms for FS in the classification task. In particular, IBAO-SVM provides a much better effect across all benchmarks

Table 9Comparisons of IBAO with a few meta-heuristic methods using SVM classifier according to the mean accuracy (Mean_{Acc}).

Benchmarks	Metric	BAO	IBAO	BSSA	BABC	BPSO	BBA	BGWO	BWOA	BGOA	BSFO	BHHO	BBSA	BASO	BHGSO
BreastEW	Mean	0.9474	0.9474	0.9447	0.9424	0.9395	0.9325	0.9444	0.9404	0.9395	0.9468	0.9433	0.9421	0.9330	0.9386
	SD	0.0000	0.0000	0.0040	0.0043	0.0026	0.0096	0.0041	0.0035	0.0047	0.0022	0.0049	0.0043	0.0083	0.0000
BreastCancer	Mean	0.9767	0.9786	0.9776	0.9786	0.9733	0.9707	0.9776	0.9760	0.9776	0.9786	0.9755	0.9767	0.9664	0.9729
	SD	0.0032	0.0000	0.0024	0.0000	0.0037	0.0065	0.0024	0.0034	0.0024	0.0000	0.0035	0.0032	0.0042	0.0034
Exactly	Mean	0.7320	0.7450	0.7085	0.7415	0.7022	0.6777	0.7187	0.7060	0.7135	0.7435	0.6973	0.7013	0.6683	0.6870
	SD	0.0264	0.0000	0.0304	0.0132	0.0311	0.0139	0.0305	0.0304	0.0318	0.0081	0.0316	0.0292	0.0051	0.0234
Exactly2	Mean	0.7500	0.7500	0.7500	0.7500	0.7500	0.7500	0.7500	0.7500	0.7500	0.7500	0.7500	0.7500	0.7500	0.7500
	SD	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CongressEW	Mean	0.9770	0.9770	0.9770	0.9770	0.9759	0.9724	0.9770	0.9766	0.9766	0.9770	0.9770	0.9766	0.9682	0.9747
	SD	0.0000	0.0000	0.0000	0.0000	0.0034	0.0056	0.0000	0.0021	0.0021	0.0000	0.0000	0.0021	0.0049	0.0046
IonosphereEW	Mean	0.9653	0.9653	0.9596	0.9592	0.9451	0.9371	0.9596	0.9535	0.9516	0.9624	0.9577	0.9502	0.9258	0.9404
	SD	0.0079	0.0079	0.0079	0.0076	0.0084	0.0125	0.0070	0.0083	0.0079	0.0066	0.0096	0.0070	0.0102	0.0079
KrVsKpEW	Mean	0.9826	0.9852	0.9791	0.9799	0.9661	0.9657	0.9807	0.9742	0.9738	0.9773	0.9781	0.9757	0.9518	0.9700
	SD	0.0031	0.0036	0.0045	0.0033	0.0087	0.0089	0.0043	0.0052	0.0038	0.0025	0.0053	0.0031	0.0173	0.0038
HeartEW	Mean	0.9068	0.9074	0.9056	0.9049	0.8840	0.8617	0.8975	0.8920	0.8920	0.9074	0.8944	0.8926	0.8358	0.8562
	SD	0.0033	0.0000	0.0056	0.0063	0.0172	0.0247	0.0124	0.0136	0.0144	0.0000	0.0137	0.0168	0.0347	0.0228
M-of-n	Mean	1.0000	1.0000	1.0000	1.0000	0.9795	0.9642	1.0000	0.9953	1.0000	1.0000	1.0000	1.0000	0.8905	0.9950
	SD	0.0000	0.0000	0.0000	0.0000	0.0523	0.0650	0.0000	0.0251	0.0000	0.0000	0.0000	0.0000	0.0718	0.0269
PenglungEW	Mean	0.8111	0.8356	0.8067	0.8022	0.8000	0.8000	0.8089	0.8044	0.8044	0.8133	0.8133	0.8022	0.7889	0.8000
	SD	0.0248	0.0333	0.0200	0.0120	0.0000	0.0172	0.0227	0.0166	0.0166	0.0267	0.0267	0.0120	0.0248	0.0000
Lymphography	Mean	0.8411	0.8556	0.8422	0.8400	0.8178	0.8056	0.8378	0.8256	0.8311	0.8511	0.8289	0.8333	0.7844	0.8111
	SD	0.0141	0.0157	0.0191	0.0133	0.0187	0.0212	0.0142	0.0186	0.0147	0.0166	0.0187	0.0122	0.0223	0.0157
SpectEW	Mean	0.8179	0.8327	0.8037	0.8173	0.7790	0.7710	0.8043	0.7951	0.8043	0.8272	0.8006	0.8062	0.7525	0.7648
	SD	0.0244	0.0101	0.0251	0.0190	0.0243	0.0260	0.0273	0.0283	0.0233	0.0100	0.0305	0.0238	0.0089	0.0173
SonarEW	Mean	0.9405	0.9532	0.9317	0.9214	0.8905	0.8746	0.9254	0.9111	0.9087	0.9302	0.9254	0.9151	0.8675	0.8825
	SD	0.0119	0.0144	0.0134	0.0152	0.0200	0.0212	0.0134	0.0173	0.0164	0.0105	0.0160	0.0170	0.0219	0.0184
Vote	Mean	1.0000	1.0000	1.0000	1.0000	0.9950	0.9906	1.0000	0.9994	0.9994	1.0000	1.0000	1.0000	0.9800	0.9933
	SD	0.0000	0.0000	0.0000	0.0000	0.0076	0.0103	0.0000	0.0030	0.0030	0.0000	0.0000	0.0000	0.0117	0.0082
Tic-tac-toe	Mean	0.9062	0.9062	0.9062	0.9062	0.8868	0.8875	0.9062	0.9028	0.9062	0.9062	0.9062	0.9062	0.8457	0.8948
	SD	0.0000	0.0000	0.0000	0.0000	0.0199	0.0261	0.0000	0.0094	0.0000	0.0000	0.0000	0.0000	0.0332	0.0128
WineEW	Mean	1.0000	1.0000	1.0000	1.0000	0.9861	0.9833	0.9991	0.9954	0.9972	1.0000	1.0000	0.9991	0.9685	0.9852
	SD	0.0000	0.0000	0.0000	0.0000	0.0139	0.0136	0.0050	0.0104	0.0083	0.0000	0.0000	0.0050	0.0199	0.0139
WaveformEW	Mean	0.8781	0.8801	0.8753	0.8738	0.8579	0.8611	0.8722	0.8694	0.8669	0.8718	0.8728	0.8696	0.8512	0.8639
	SD	0.0024	0.0030	0.0041	0.0028	0.0052	0.0079	0.0037	0.0045	0.0049	0.0028	0.0045	0.0047	0.0075	0.0051
Zoo	Mean	1.0000	1.0000	1.0000	1.0000	0.9984	0.9873	1.0000	0.9984	1.0000	1.0000	0.9984	1.0000	0.9683	0.9921
	SD	0.0000	0.0000	0.0000	0.0000	0.0085	0.0211	0.0000	0.0085	0.0000	0.0000	0.0085	0.0000	0.0224	0.0177
Score	W T L	0 9 9	7 11 0	0 7 11	0 8 10	0 1 17	0 1 17	0 6 12	0 1 17	0 4 14	0 9 9	0 6 12	0 5 13	0 1 17	0 1 17

in terms of mean accuracy results, followed by the IBAO- k -NN. Moreover, the IBAO optimizer showed a more significant exploration and exploitation behavior than its peers, verified by subsequent underlying causes.

Firstly, to handle the FS in the classification task, the basic continuous variant of the AO optimizer was mapped to a binary version. Secondly, the AO algorithm has stable exploration and exploitation abilities, which were then improved by integrating the random position amendment approach and the LS strategy into its original version to avoid local optima within the solution region through iterations. Then, k -NN and SVM classifiers were used to verify the effectiveness of the suggested IBAO optimizer in dealing with FS in supervised classification against its peers using eighteen benchmarks with multi-scale features and instances obtained from the UCI machine learning repository.

Moreover, the suggested models (IBAO- k -NN and IBAO-SVM) additionally emphasize their behavior to obtain the optimal solution on time, ensuring an effective equilibrium between exploration and exploitation capabilities. Eventually, it can be shown that, due to the non-exact repeatability of the optimization outcomes, separate implementations of the optimizer can generate various subsets of attributes, which may confuse the user. Therefore, on different occasions or applications, IBAO or other optimizers implemented here can select multiple subsets of features.

5. Conclusion and future directions

This paper proposed a novel algorithm for handling FS strategies that incorporated the AO optimizer, the random position amendment

approach, and the LS strategy to enhance the optimal results achieved after each algorithm iteration. Satisfactory results have been achieved, revealing how the performance and capability of the algorithm have been greatly improved. Two standard classifiers k -NN and SVM, were used to calculate the classification accuracy for each subset of the chosen features. The suggested IBAO algorithm on multiple benchmarks with multi-scale features and instances was verified by comparing its results with binary variants of 13 state-of-art meta-heuristic methods. The evaluation was performed using various performance metrics, including mean fitness rate, mean accuracy rate, and the mean number of features selected. By analyzing the experimental results, the two algorithms presented in this paper (i.e., IBAO- k -NN and IBAO-SVM) achieved more reliable results than their counterparts in handling FS strategies. In particular, IBAO-SVM provided a much better effect across all benchmarks regarding mean accuracy results, followed by IBAO- k -NN. Moreover, the IBAO optimizer showed more significant exploration and exploitation behavior than its peers. Eventually, according to the Wilcoxon's test ($\alpha = 0.05$), the dominance of the IBAO algorithm with k -NN and SVM classifiers was evident in comparison with its peers and obtained classification accuracy up to 100% in some of these benchmarks integrated with a feature reduction size down to 92%.

The combination of the AO algorithm and various meta-heuristic algorithms may be investigated for future work. In addition, other classification methods (such as neural networks, decision trees, random forest, etc.) can be used to further verify the behavior of the IBAO algorithm in FS in classification tasks.

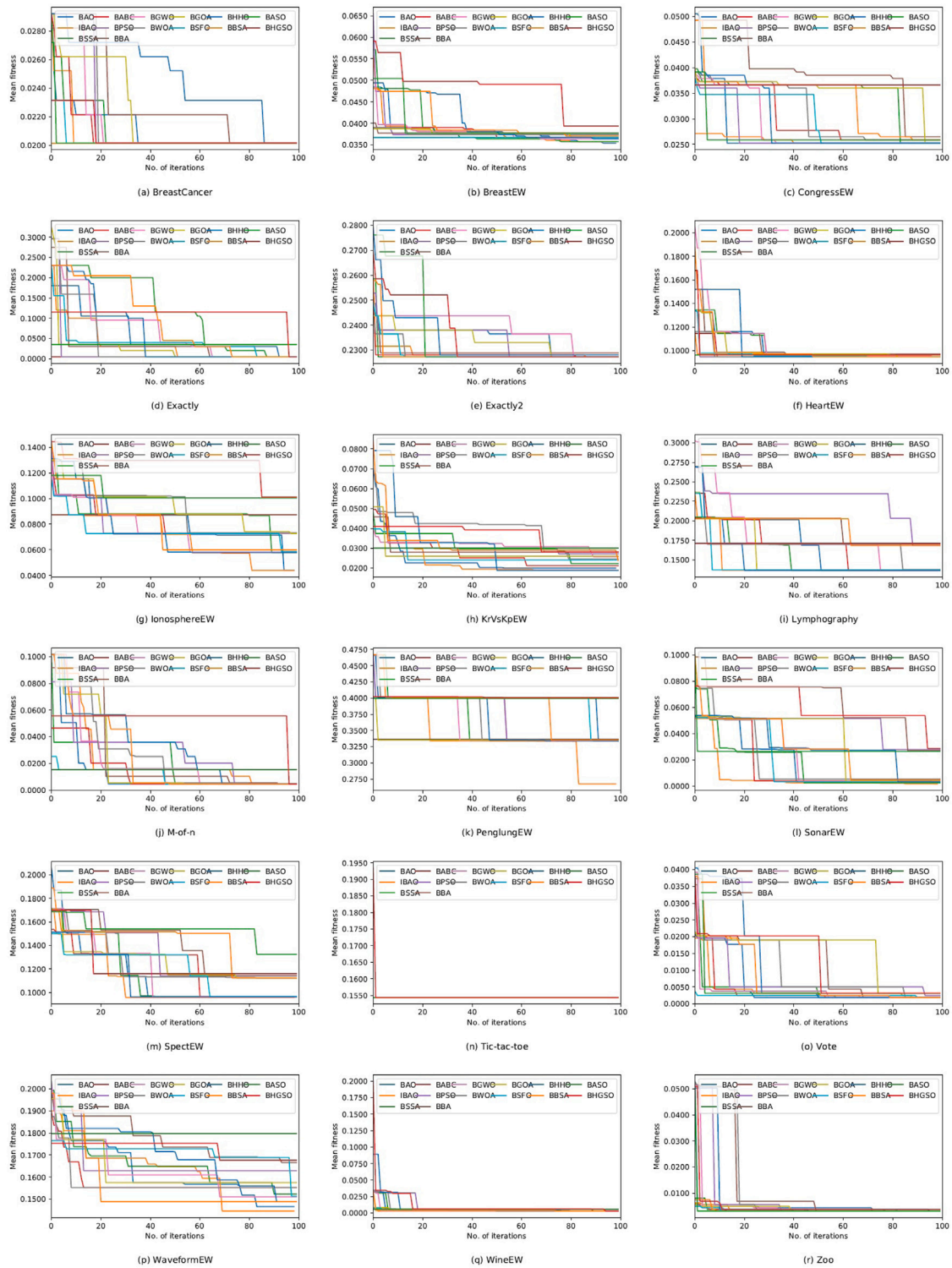


Fig. 2. Convergence graphs of the suggested IBAO and other methods with the k -NN algorithm over all benchmarks.

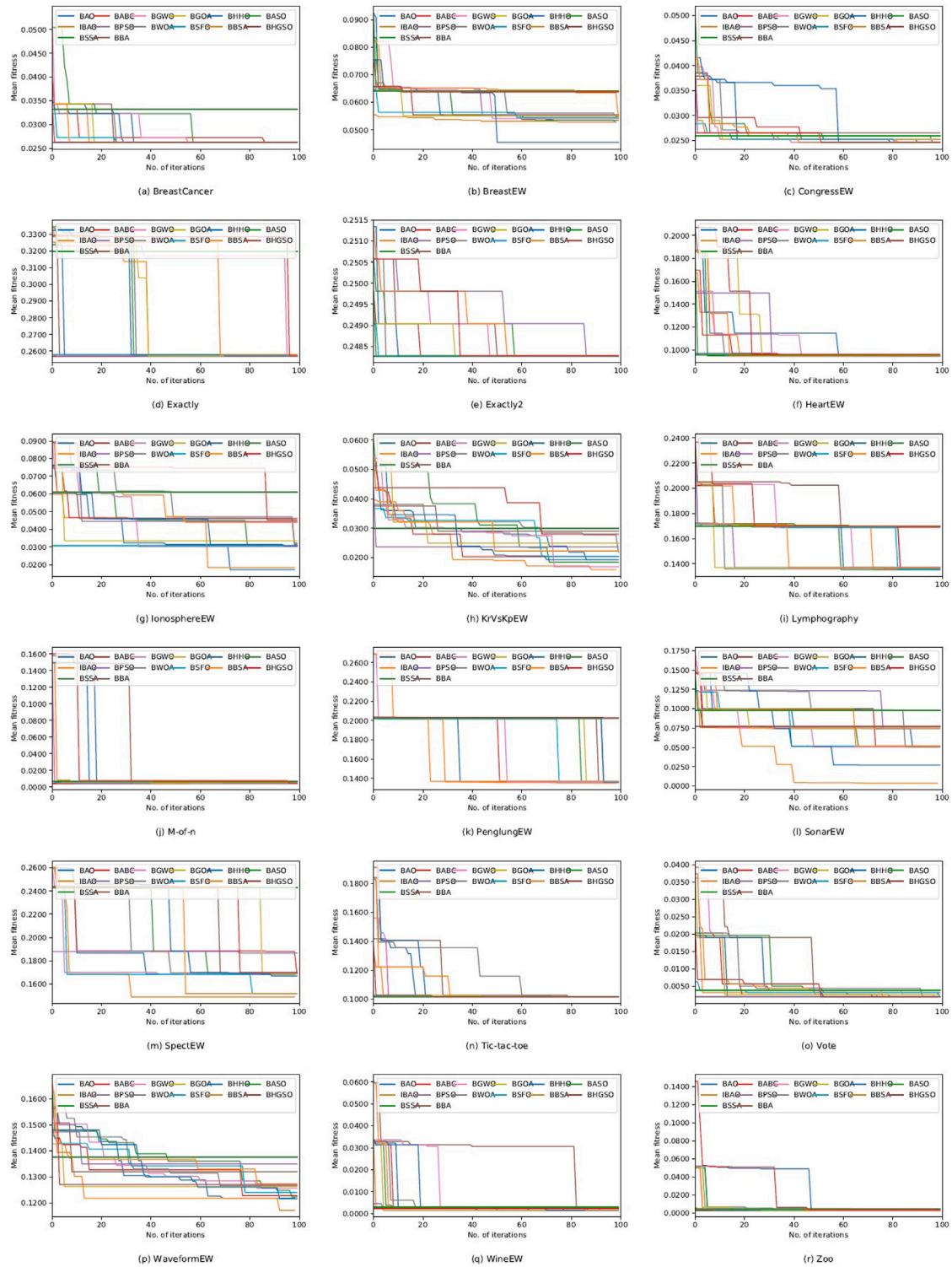


Fig. 3. Convergence graphs of the suggested IBAO and other methods with the SVM algorithm over all benchmarks.

Table 10

Comparisons of IBAO with a few meta-heuristic methods using SVM classifier according to the mean number of features selected (Mean_{Feat}).

Benchmarks	Metric	BAO	IBAO	BSSA	BABC	BPSO	BBA	BGWO	BWOA	BGOA	BSFO	BHHO	BBSA	BASO	BHGSO
BreastEW	Mean	007.10	005.77	007.97	010.70	011.87	012.80	009.43	010.10	010.03	010.17	009.13	010.23	013.53	013.93
	SD	002.02	001.61	001.82	001.27	001.50	002.66	001.63	001.94	001.25	001.46	001.78	001.43	002.81	002.22
BreastCancer	Mean	004.77	005.10	005.07	005.10	004.97	004.93	004.97	004.93	005.20	005.00	004.83	005.00	004.97	005.10
	SD	000.50	000.30	000.57	000.30	000.91	000.89	000.48	000.68	000.65	000.00	000.69	000.52	001.40	000.75
Exactly	Mean	006.33	006.30	007.07	006.87	007.53	007.40	006.53	007.60	007.27	006.43	006.23	007.17	004.83	008.03
	SD	001.35	000.46	001.59	000.85	001.67	002.32	001.54	001.58	001.59	000.67	002.69	001.49	002.65	001.62
Exactly2	Mean	001.00	001.00	001.03	001.43	002.27	002.47	001.03	001.50	001.30	001.00	001.00	001.53	003.13	002.80
	SD	000.00	000.00	000.18	000.50	000.63	000.76	000.18	000.62	000.46	000.00	000.00	000.50	000.81	000.79
CongressEW	Mean	003.93	003.47	004.57	005.33	007.13	007.20	004.47	006.10	005.57	004.20	004.97	004.87	007.00	008.03
	SD	000.81	000.62	000.96	001.01	002.06	002.34	000.92	001.70	001.50	000.65	001.14	001.06	001.57	002.37
IonosphereEW	Mean	011.73	011.33	012.87	015.87	015.47	016.27	014.13	014.93	015.67	013.63	013.30	013.87	016.60	017.37
	SD	001.93	002.51	002.29	002.49	003.04	003.03	003.29	002.73	002.65	002.29	002.73	003.14	003.08	003.23
KrVsKpEW	Mean	021.90	022.50	022.43	024.03	021.47	022.73	023.47	021.40	023.20	021.57	021.87	022.70	022.13	023.37
	SD	002.91	002.92	003.07	002.59	003.69	003.55	003.61	003.53	003.13	002.87	003.15	002.79	002.77	002.74
HeartEW	Mean	004.37	004.07	004.50	005.00	005.00	004.67	005.07	005.13	005.00	004.63	005.07	004.77	005.60	005.33
	SD	000.48	000.25	000.76	001.00	001.18	001.49	001.03	000.99	001.00	000.55	001.09	000.96	001.56	001.16
M-of-n	Mean	006.00	006.07	006.50	006.83	008.13	008.40	006.53	007.23	007.27	006.57	006.73	007.30	008.43	008.67
	SD	000.00	000.25	000.62	000.58	001.41	001.62	000.62	000.88	000.57	000.50	000.73	000.82	001.76	000.98
PenglungEW	Mean	125.20	102.83	126.87	150.67	152.37	151.20	134.33	142.80	143.90	133.17	126.37	142.73	155.13	167.57
	SD	008.80	010.44	007.51	008.10	007.89	014.75	006.86	007.01	005.78	013.22	006.28	006.14	014.26	003.94
Lymphography	Mean	007.63	007.87	008.13	008.83	008.57	009.07	008.70	008.33	009.67	007.93	008.37	008.87	009.80	009.73
	SD	001.38	000.99	001.89	001.07	002.04	002.71	001.37	002.01	002.34	001.31	001.30	001.65	002.06	001.97
SpectEW	Mean	007.57	007.60	009.20	010.03	009.97	011.57	008.83	009.40	009.70	009.43	007.70	009.40	010.63	011.70
	SD	001.91	002.09	002.06	001.14	002.09	002.11	001.86	002.23	001.55	001.50	002.45	001.47	003.65	001.53
SonarEW	Mean	023.77	022.10	025.10	030.20	028.90	029.53	026.97	026.83	028.30	027.63	025.67	027.83	030.00	031.67
	SD	002.88	003.13	003.56	004.00	004.32	004.38	003.34	003.53	003.26	003.72	004.22	002.85	004.11	004.68
Vote	Mean	003.77	003.40	004.00	005.20	006.47	006.73	004.73	005.83	005.93	004.17	005.07	005.33	007.27	006.63
	SD	000.80	000.66	001.13	001.28	001.73	001.93	001.24	001.66	001.12	000.93	001.29	001.60	001.69	001.66
Tic-tac-toe	Mean	008.00	008.00	008.00	008.00	007.67	007.53	008.00	008.00	008.13	008.00	008.13	008.03	006.63	007.87
	SD	000.00	000.00	000.00	000.00	000.83	000.92	000.00	000.45	000.34	000.00	000.34	000.18	001.14	000.85
WineEW	Mean	002.27	002.17	002.90	003.53	004.77	005.17	003.13	003.77	004.20	002.47	003.40	003.70	005.70	005.50
	SD	000.63	000.45	001.16	000.92	001.23	001.42	000.96	001.05	001.19	000.50	001.14	001.22	001.24	001.50
WaveformEW	Mean	024.23	023.20	024.17	026.30	024.37	024.10	024.80	025.53	024.93	025.57	024.20	024.50	025.27	025.53
	SD	002.50	002.39	002.96	002.35	002.76	003.17	002.98	002.23	002.62	002.72	002.29	002.55	003.47	003.01
Zoo	Mean	005.23	005.13	005.80	006.03	007.77	007.07	006.07	006.30	006.93	005.40	005.80	006.30	007.53	008.03
	SD	000.50	000.34	001.17	000.88	001.65	001.95	000.93	001.07	001.24	000.55	000.91	001.22	001.82	001.76
Score	W T L	4 1 13	10 1 7	0 0 18	0 0 18	0 0 18	0 0 18	0 0 18	1 0 17	0 0 18	0 1 17	0 1 17	0 0 18	2 0 16	0 0 18

Table 11

p-values of the Wilcoxon's test for the mean fitness results of the IBAO-*k*-NN against other algorithms.

IBAO- <i>k</i> -NN vs.	<i>R</i> ⁺	<i>R</i> ⁻	<i>P</i> -value	Winner
BAO	169.5	1.5	1.9069E-5	IBAO
BSSA	171.0	0.0	7.63E-6	IBAO
BABC	150.0	3.0	7.63E-5	IBAO
BPSO	171.0	0.0	7.63E-6	IBAO
BBA	171.0	0.0	7.63E-6	IBAO
BGWO	171.0	0.0	7.63E-6	IBAO
BWOA	171.0	0.0	7.63E-6	IBAO
BGOA	171.0	0.0	7.63E-6	IBAO
BSFO	146.5	24.5	0.006088	IBAO
BHHO	171.0	0.0	7.63E-6	IBAO
BBSA	171.0	0.0	7.63E-6	IBAO
BASO	171.0	0.0	7.63E-6	IBAO
BHGSO	171.0	0.0	7.63E-6	IBAO

Table 12

p-values of the Wilcoxon's test for the mean fitness results of the IBAO-SVM against other algorithms.

IBAO-SVM vs.	<i>R</i> ⁺	<i>R</i> ⁻	<i>P</i> -value	Winner
BAO	147.5	5.5	1.8309E-4	IBAO
BSSA	169.5	1.5	1.9069E-5	IBAO
BABC	169.5	1.5	1.9069E-5	IBAO
BPSO	171.0	0.0	7.63E-6	IBAO
BBA	171.0	0.0	7.63E-6	IBAO
BGWO	169.5	1.5	1.9069E-5	IBAO
BWOA	171.0	0.0	7.63E-6	IBAO
BGOA	171.0	0.0	7.63E-6	IBAO
BSFO	166.5	4.5	6.485E-5	IBAO
BHHO	153.0	0.0	1.5258E-5	IBAO
BBSA	153.0	0.0	1.5258E-5	IBAO
BASO	171.0	0.0	7.63E-6	IBAO
BHGSO	171.0	0.0	7.63E-6	IBAO

CRediT authorship contribution statement

Amr A. Abd El-Mageed: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Validation, Writing – original draft, Writing – review & editing. **Amr A. Abohany:** Resources, Formal analysis, Data curation, Validation, Writing – original draft, Writing – review & editing. **Ahmed Elashry:** Investigation, Visualization, Data curation, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Table 13

Comparison of IBAO and other methods from the literature according to the average accuracy.

Benchmarks	IBAO-k-NN	IBAO-SVM	TCSSA3 (Aljarah et al., 2018)	BGOA-EPD (Mafarja et al., 2018)	GA (Emary et al., 2016)	BSSA-CP (Faris et al., 2018)
BreastEW	0.9649	0.9474	N/A	0.9470	0.9390	0.9484
BreastCancer	0.9857	0.9786	N/A	0.9800	0.9680	0.9768
Exactly	1.0000	0.7450	0.9969	0.9990	0.6740	0.9803
Exactly2	0.7730	0.7500	0.7672	0.7800	0.7460	0.7582
CongressEW	0.9770	0.9770	0.9705	0.9640	0.9320	0.9628
IonosphereEW	0.9362	0.9653	0.9377	0.8990	0.8140	0.9182
KrVsKpEW	0.9824	0.9852	0.9692	0.9680	0.9200	0.9644
HeartEW	0.9074	0.9074	0.8331	0.8330	0.7800	0.8605
M-of-n	1.0000	1.0000	0.9992	1.0000	0.8610	0.9918
PenglungEW	0.6533	0.8356	0.9072	0.9270	0.5840	0.8775
Lymphography	0.8444	0.8556	0.8444	0.8680	0.6960	0.8900
SpectEW	0.8920	0.8327	0.8333	0.8260	0.7930	0.8361
SonarEW	0.9897	0.9532	0.9481	0.9120	0.7540	0.9372
Vote	1.0000	1.0000	0.9549	0.9660	0.9040	0.9511
Tic-tac-toe	0.8542	0.9062	0.7975	0.8080	0.7190	0.8205
WineEW	1.0000	1.0000	0.9978	0.9890	0.9370	0.9933
WaveformEW	0.8501	0.8801	0.7364	0.7370	0.7730	0.7335
Zoo	1.0000	1.0000	0.9928	0.9930	0.8550	1.0000
Score (W T L)	5 6 7	4 6 8	0 0 18	2 1 15	0 0 18	1 1 16

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